

Unit Functional Design Specification - Hydrogenation 1

Wilmar Industrial Complex PT MNA Serang, Indonesia

End Customer: Wilmar Industrial Complex PT MNA Serang,
Indonesia

End Customer Reference Number: -

Document File Name: 3258585-PSS-PAS-FSP-00010

Document Status: For Final

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Revision History

Rev	Status	Description	Date	Developed By	Reviewed By
A	For Approval	Issue for Approval	26/10/2018	SJ	AS
0	For Final	Issue for Final	14/10/2019	DG	AS

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1 EQUIPMENT MODULE CLASS DESIGN – EM_HY_STARTUP (1-1300 STARTUP)

1.1 Functional Description

EM_HY_STARTUP is a class based equipment module that drives six other sequences to complete hydrogenation for different products as per the selected jobs. EM_HY_STARTUP is derived from the Command Driven equipment module class _EM_CMD32_130 which is defined in a separate document. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Aux On Command and It shall be released in Aux Off Command. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as buttons on graphics wherever possible.

1.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_HY_STARTUP	HY1EQ_1300
DEV1	_MRSS11_130_SE	HY1_PU1078TW
DEV2	_V12_IP_130_SE	HY1_V131078TW
DEV3	EM_HY_DERTN_PROD	HY1EQ_1303_82A

Table 1: Instances

1.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

1.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Common Hold Start	A0010	Store Fail Message
		A0020	Reset HOLD_REQ
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Holding DEV3

Step	Description	Action	Action Description
T9900A	Actions Complete or Time Out & Hold=True		Go to step H9900A
T9900B	Actions Complete or Time Out & Hold=False		Go to step H9900B
H9900A	Common Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Reset Parameters
END1	Command Complete		True

Table 2: Hold Sequence

1.3.2 Command 01 – Aux On

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Aux On Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initilizations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Set Task Pointer
		A0040	Acquire EM
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Reset Hold Request
T0100	Standby		Go to step S0100
S0100	Standby	A0010	Stopping and Disabling Conditional Logic for DEV1



Step	Description	Action	Action Description
		A0020	Closing and Disabling Conditional Logic for DEV2
T0200	Start Aux		If no pending actions in S0100 then go to step S0200
S0200	Start Aux	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Waiting for LSL_078TW to be normal
T9900	Complete		If LSL_078TW is not in alarm then go to step S9900
S9900	Complete	A0010	Set Message: Aux On Complete
END	Command Complete		True

Table 3: Aux On Sequence

1.3.3 Command 02 – Aux Off

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Aux Off Command
T0010	Reset Hold Request		Go to step S0010
S0010	Reset Hold Request	A0010	Reset Hold Request
T0100	Aux Off		If no pending actions in S0010 then go to step S0100
S0100	Aux Off	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Set/Reset Buttons
T0200	Release Devices		If no pending actions in S0100 then go to step S0200
S0200	Release Devices	A0010	Set Message: Disable HM and SM
		A0020	Release Private Device
		A0030	Release Private Device
T9900	Actions Complete		Go to step S9900
S9900	Complete	A0010	Release Devices Message
		A0020	Set Message: Aux Off Complete
END	Command Complete		True



Table 4: Aux Off Sequence**1.3.4 Command 03 – Change Feed**

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Change Feed Command
T0010	Actions Complete		If Task Pointer = 1 then go to step S0010
T0100B	Validation Check		If Task Pointer = 2 then go to step S0100
T0200B	Shift Jobs		If Task Pointer = 3 then go to step S0200
T0300B	Production		If Task Pointer = 4 then go to step S0300
S0010	Reset Hold Request	A0010	Reset Hold Request
T0100A	Validation Check		If no pending actions in S0010 then proceed step S0100
S0100	Validation Check	A0010	Waiting for Oil Dearth and Vacuum Production to be in standby
		A0020	Set Dynamic References
		A0030	Checking Data Validation
		A0040	Set Task Pointer
		A0050	Enabling Conditional Logic for DEV1
		A0060	Enabling Conditional Logic for DEV2
T0200	Shift Jobs		Job-1 List Parameters Validation OK then proceed to step S0200
S0200	Shift Jobs	A0010	Set Dynamic References
		A0020	Shifting Jobs
		A0030	Resetting Batch No.
		A0040	Setting Batches No.
		A0050	Disable the Change Feed Button
		A0060	Set Task Pointer
		A0070	Enabling Conditional Logic for DEV1
		A0080	Enabling Conditional Logic for DEV2
T0300	Production		If no pending actions in S0200 then proceed step S0300
S0300	Production	A0005	Waiting for Oil Dearth and Vacuum Production to be in standby
		A0010	Set Message: Production is in progress
		A0020	Initiate Command to Other EM (DEV3)
		A0030	Waiting for Required Batch = Finished Batch
		A0040	Set Task Pointer

Step	Description	Action	Action Description
		A0050	Enabling Conditional Logic for DEV1
		A0060	Enabling Conditional Logic for DEV2
T0100C	Validation Check		If Required Batches (OP001) = Actual Batches (TP003) and Operator select <Continue> command then go to step S0100
T300B	Production		If Required Batches (OP001) < Actual Batches (TP003) then go to step S0300
T9900C	Actions Complete		If Operator select <Drain> command (OP004) then go to step S9900
S9900	Complete	A0010	Production Command Complete
END	Command Complete		True

Table 5: Change Feed Sequence

1.3.5

Command 04 – Drain

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Drain Command
T0010	Actions Complete		Go to step S0010
S0010	Reset Hold Request	A0010	Reset Hold Request
		A0020	Reset Buttons (OP004) & (OP005)
T0100	Actions Complete		Go to step S0100
S0100	Drain	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Set Message: Waiting for Other EM's to Complete
T9900	Actions Complete		Go to step S9900
S9900	Complete	A0010	Set Message: Drain Command Complete
END	Command Complete		True

Table 6: Drain Sequence

1.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value / Reference
Operating Parameters				
OP001	Required Batches	Floating Point	NA	0
OP002	AUX	Name Set	NA	OFF
OP003	HOLD	Name Set	NA	OFF
OP004	DRAIN	Name Set	NA	OFF
OP005	RUN	Name Set	NA	OFF
OP006	Tank Farm Ready	Boolean	NA	False
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Not Used	Floating Point	NA	0
TP003	Actual Batches	External Reference	NA	HY1_RECIP/ACTUAL_BATCH_7

Table 7: Operating and Tuning Parameters

1.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value / Reference
MY_TAG		HY1EQ_1300
OPR_CMD_MASK		31
CAT_DOSING_A	External Reference	HY1EQ_1335A/OWNER_ID
REACT_FILT	External Reference	HY1EQ_1382C/ OWNER_ID
REACTOR1	External Reference	HY1EQ_1303/ OWNER_ID
REACTOR2	External Reference	HY1EQ_1382B/ OWNER_ID
CAT_DOSING_B	External Reference	HY1EQ_1335B/ OWNER_ID
LSL_078TW	External Reference	HY1_LSL1078TW/DI1/DISC_ACT
RECIPE	String	HY1_RECEIPE
DYN_REF_1	Dynamic Reference	
CONTINUE	Boolean	
CHANGE_FEED	Boolean	

Table 8: Configurable Parameters

1.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_HY_STARTUP	Hydrogenation Area: Startup EM

Table 9: Named Set Details

Details of states for NS_HY_STARTUP named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Aux ON
02	Aux OFF
03	Change Feed
04	Drain

Table 10: Name Set State Details

2 EQUIPMENT MODULE CLASS DESIGN – EM_HY_DERTN_PROD (1-1302/82A)

2.1 Functional Description

EM_HY_DERTN_PROD is a class based equipment module that performs filling operation and later heats the material to a defined temperature. EM_HY_DERTN_PROD is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Heat Command and shall be released on completion of Heat Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

2.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_HY_DERTN_PROD	HY1EQ_1303_82A
DEV1	_V12_IP_130	HY1_V1001F
DEV2	_V12_IP_130	HY1_V1001BN
DEV3	_V12_IP_130	HY1_V1002F
DEV4	_V12_IP_130	HY1_V1382AMPS
DEV5	_V12_IP_130	HY1_V1002LPS
DEV6	_CSD_130_SE	HY1_FT1001
DEV7	Ignored	Ignored
DEV8	EM_HY_CAT_DOSING	HY1EQ_1335A
DEV9	EM_HY_CAT_DOSING	HY1EQ_1335B
DEV10	_O_TOT_130_SE	HY1_FQ1001
DEV11	EM_HY_REACTOR1	HY1EQ_1303

Table 11: Instances

2.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

2.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time

Step	Description	Action	Action Description
T1000	Actions Complete		Go to step H1000
H1000	Hold	A0010	Set Message: Setting Devices
		A0020	Closing DEV1
		A0030	Closing and Disabling Conditional Logic for DEV2
		A0040	Closing DEV3
		A0050	Closing and Disabling Conditional Logic for DEV4
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Set DEV6 Mode to Remote Out
		A0080	Set DEV6 output to Closed
T9900A	Actions Complete or Time Out & Hold=True		Go to step H9900A
T9900B	Actions Complete or Time Out & Hold=False		Go to step H9900B
H9900A	Reset	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	Release Private Device
		A0100	Reset Parameters
END1	Command Complete		True

Table 12: Hold Sequence

2.3.2 Command 01 – Heat

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Staring Heat Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	Reset Totalizer
		A0030	First Pass = True
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Device	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Acquire Private Device
		A0060	Acquire Private Device
		A0070	Acquire Private Device
		A0080	Acquire Private Device
		A0090	Acquire Private Device
		A0100	Reset Hold Request
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0400B	Fill		If Task Pointer = 2 then go to step S0400
T0500B	Blow Inlet		If Task Pointer = 3 then go to step S0500
T0600B	Heat		If Task Pointer = 4 then go to step S0600
T0700B	Ready		If Task Pointer = 5 then go to step S0700
T0800B	Discharge		If Task Pointer = 6 then go to step S0800
T0900B	Drain Wait		If Task Pointer = 7 then go to step S0900
S0100	Standby	A0010	Closing DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Set DEV6 Mode to Remote Out
		A0070	Set DEV6 output to Closed
T0200	EM Selection (1335A or B)		If EM_1335A is in <Standby> then go to step S0200

Step	Description	Action	Action Description
S0200	EM Selection (1335A or B)	A0010	Acquiring Catalyst EM Dosing based on selection (EM_1335B Selection)
T0300	Set Device		If no pending actions in S0200 then go to step S0300
S0300	Set Device	A0010	Initiate Command to EM_1335A
		A0020	Initiate Command to EM_1335B (If it is selected by Operator)
		A0040	Checking for LSH_82C to not be in Alarm
T0400A	Fill		If EM_STARTUP is in <Production> and not in <Hold> command and operator command <Fill> (OP004) is selected and XV 1-1303-F closed feedback is available and ((EM_1382B Product (TP001 of EM_1382B) = EM_1382C Product (TP017 of EM_1382C) and LSH_82C not in alarm) or EM_1382B is in <Standby>) then go to step S0400
S0400	Fill	A0010	Copy Product Code and Increment Batch Number (TP006) and Set Task Pointer
		A0020	Opening DEV1
		A0030	Closing and Disabling Conditional Logic for DEV2
		A0040	Opening DEV3
		A0050	Enabling Conditional Logic for DEV4
		A0060	Enabling Conditional Logic for DEV5
		A0070	Set Totalizer
		A0080	Set DEV6 Mode to Remote Cascade and Start Totalizer
		A0090	Set DEV6 setpoint to Loop Setpoint for DEV6 and Enabling Conditional Logic
		A0100	Message to wait for Totalizer
T0500A	Blow Inlet		If (DEV10 (Tons) >= Quantity Required Setpoint (Batch Size) or LSH_002A is in alarm then go to step S0500
S0500	Blow Inlet	A0010	Closing DEV1 and Set Task Pointer
		A0020	Enabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Set DEV6 Mode to Remote Out
		A0070	Set DEV6 output to Open and Disabling Conditional Logic

Step	Description	Action	Action Description
		A0080	Set Blow Inlet Time
		A0090	Start Blow Inlet Time
		A0100	Blow Inlet Time Message
T0600A	Heat		If Blow time (TP002) is over then go to step S0600
S0600	Heat	A0010	Set Task Pointer
		A0020	Set Scratch Timer and Temperature Setpoint
		A0030	Closing DEV1
		A0040	Closing and Disabling Conditional Logic for DEV2
		A0050	Closing DEV3
		A0060	Enabling Conditional Logic for DEV4
		A0070	Enabling Conditional Logic for DEV5
		A0080	Set DEV6 Mode to Remote Out and Disabling Conditional Logic for DEV6
		A0090	Set DEV6 output to Closed
		A0100	Waiting for Delay Timer and Temperature Setpoint to complete
T0700A	Ready		If TT_002A >= Setpoint Heating Temperature (TP004) and Wait time (TP003) is over then go to step S0700
S0700	Ready	A0010	Set Task Pointer
		A0020	Starting Reactor 1
		A0030	Closing DEV1
		A0040	Closing and Disabling Conditional Logic for DEV2
		A0050	Closing DEV3
		A0060	Enabling Conditional Logic for DEV4
		A0070	Enabling Conditional Logic for DEV5
		A0080	Set DEV6 Mode to Remote Out and Disabling Conditional Logic for DEV6
		A0090	Set DEV6 output to Closed
		A0100	Set Message: Waiting for EM_HY_REACTOR1 to be in step
T0800A	Discharge		If EM-1-1303 is in <Fill> and LSL_203 is in alarm then go to step S0800
S0800	Discharge	A0010	Closing DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4

Step	Description	Action	Action Description
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Set DEV6 Mode to Remote Out and Disabling Conditional Logic for DEV6
		A0070	Set DEV6 output to Closed
		A0080	Set Message: Waiting for level to be in alarm and valve to be open
		A0090	Set Scratch Timer
T0900A	Drain Wait		If ((V_203F Open feedback is available & LSL_203 is in alarm + 5 sec delay) or LSH_303 is in alarm) then go step S0900
S0900	Drain Wait	A0010	Set Task Pointer
		A0020	Closing DEV1
		A0030	Closing and Disabling Conditional Logic for DEV2
		A0040	Closing DEV3
		A0050	Closing and Disabling Conditional Logic for DEV4
		A0060	Closing and Disabling Conditional Logic for DEV5
		A0070	Set DEV6 Mode to Remote Out and Disabling Conditional Logic for DEV6
		A0080	Set DEV6 output to Closed
		A0090	Set Message: Waiting for EM_HY_REACTOR1 to complete step
T1000	Disable HM and SM		If EM-1-1303 is not in <Fill> & V_203F close feedback is available then go to step S1000
S1000	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T1100	Release Devices		If no pending actions in S1000 then go to step S1100
S1100	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
T9900	Complete		If no pending actions in S1100 then go to step S9900
S9900	Complete	A0010	Set Message: Heat Command Complete
		A0020	Self Release the EM & Reset First Pass

Step	Description	Action	Action Description
END	Command Complete		True

Table 13: Heat Sequence

2.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	MODE	Boolean	NA	MAN = 0 (AUTO = 1)
OP002	RESET	Boolean	NA	False
OP003	1335B	Boolean	NA	False
OP004	FILL	Boolean	NA	False
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Blow Inlet time	Floating Point	sec	20
TP003	Wait Fixed Timers	Floating Point	sec	30
TP004	Heating Temperature SP	External Reference	°C	HY1_RECIP/HT_TP_1302_1
TP005	Product Code	String	NA	5
TP006	Batch Number	Floating Point	NA	0
TP007	Batch Size	String	NA	15
TP008	RSP for FT_001	Floating Point	kg/h	0

Table 14: Operating and Tuning Parameters

2.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		HY1EQ_1303_82A
OPR_CMD_MASK		3
REACTOR1	String	HY1EQ_1303
TT_002A	String	HY1_TT1002A
LSL_203	External Reference	HY1_LSL1203/DI1/DISC_ACT
V_203F	External Reference	HY1_V1203F/PV_D

Parameter Name	Parameter Type	Value/Reference
LSL_002A	External Reference	HY1_LSL1002A/DI1/DISC_ACT
LSH_002A	External Reference	HY1_LSH1002A/DI1/DISC_ACT
DY_REF_1	Dynamic Reference	
DY_REF_2	Dynamic Reference	
HYXEQ-1300	String	
EM-1-1303	String	
EM_1335A	String	HY1EQ_1335A
EM_1335B	String	HY1EQ_1335B
RECIPE	String	HY1_RECEIPE
LSH_82C	External Reference	HY1_LSH1282C1/DI1/DISC_ACT
STEP_TIME	Floating Point	0
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False
EM_1382B	String	HY1EQ_1382B
EM_1382C	String	HY1EQ_1382C
PR_CODE	String	HY1_RECEIPE
HOLD	External Reference	HY1EQ_1300/OP003_VALUE
EM_STARTUP	External Reference	HY1EQ_1300/COMMAND_00003/ S0300/ACTIVE
LSH_303	External Reference	HY1_LSH1203/DI1/DISC_ACT

Table 15: Configurable Parameters

2.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_HY_DERTN_PROD	Hydrogenation Area: Oil Dearth and Vacuum Production

Table 16: Named Set Details

Details of states for NS_HY_DERTN_PROD named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Heat

Table 17: Name Set State Details

3 EQUIPMENT MODULE CLASS DESIGN – EM_HY_REACTOR1 (1-1303)

3.1 Functional Description

EM_HY_REACTOR1 is a class based equipment module. EM_HY_REACTOR1 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All Commands shall be available to operator. All devices used in EM shall be acquired in Hydrogenation Command and shall be released in Drain Command. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

3.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_HY_REACTOR1	HY1EQ_1303
DEV1	_MRSS11_130_SE	HY1_AG1203
DEV2	_V12_IP_130	HY1_V1203F
DEV3	_V12_IP_130	HY1_V1203D
DEV4	_V12_IP_130	HY1_V1203VA
DEV5	_V12_IP_130	HY1_V1203V1
DEV6	Ignored	Ignored
DEV7	_CMT_130_SE	HY1_PT1203
DEV8	_CSD_130_SE	HY1_FT1203H
DEV9	_V12_IP_130	HY1_V1203H
DEV10	_V12_IP_130	HY1_V1235
DEV11	_O_TOT_130_SE	HY1_FQ1203H

Table 18: Instances

3.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

3.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Start	A0010	Store Fail Message
		A0020	Reset Hold Request

Step	Description	Action	Action Description
T1000	Actions Complete or Time Out & Hold=True		Go to step H1000
T2000	Actions Complete or Time Out & Hold=False		Go to step H2000
H1000	Hold	A0010	Stopping DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Set DEV8 Mode to Remote Out
		A0080	Set DEV8 output to Closed and Disable Conditional Logic
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Reset DEV11
T9900A	Actions Complete or Time Out		Go to step H9900A
H2000	General Reset	A0010	Stopping DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Set DEV8 Mode to Remote Out
		A0080	Set DEV8 output to Closed and Disable Conditional Logic
		A0090	Closing and Disabling Conditional Logic for DEV9

Step	Description	Action	Action Description
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Reset DEV11
T9900B	Actions Complete or Time Out		Go to step H9900B
H9900A	Hold Complete	A0010	Completed
END	Command Complete		True
H9900B	Release All Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	Release Private Device
		A0100	Release Private Device
		A0110	Release Private Device
		A0120	Release Private Device
		A0130	Reset Parameters
END1	Command Complete		True

Table 19: Hold Sequence

3.3.2 Command 01 – Hydrogenation

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Hydrogenation Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Reset Totalizer
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020

Step	Description	Action	Action Description
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Acquire Private Device
		A0060	Acquire Private Device
		A0070	Acquire Private Device
		A0080	Acquire Private Device
		A0090	Acquire Private Device
		A0100	Acquire Private Device
		A0110	Acquire Private Device
		A0120	Acquire Private Device
		A0130	Reset Hold Request
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0200B	Fill		If Task Pointer = 2 then go to step S0200
T0300B	Heating		If Task Pointer = 3 then go to step S0300
T0400D	CatalystHYD		If Task Pointer = 4 then go to step S0400
T0500C	Parameter Check		If Task Pointer = 5 then go to step S0500
T0600C	ReactionHYD		If Task Pointer = 6 then go to step S0600
T0700B	Reaction Fail		If Task Pointer = 7 then go to step S0700
T0800C	Selection HYD		If Task Pointer = 8 then go to step S0800
S0100	Standby	A0020	Reset Product Code & Batch Number
		A0030	Stopping DEV1
		A0040	Closing and Disabling Conditional Logic for DEV2
		A0050	Closing and Disabling Conditional Logic for DEV3
		A0060	Closing and Disabling Conditional Logic for DEV4
		A0070	Enabling Conditional Logic for DEV5
		A0080	Closing and Disabling Conditional Logic for DEV6
		A0090	Set DEV8 Mode to Remote Out
		A0100	Set DEV8 output to Closed and Disable Conditional Logic
		A0110	Closing and Disabling Conditional Logic for DEV9
		A0120	Closing and Disabling Conditional Logic for DEV10
		A0130	Wait Message
T0200A	Fill		If DERTN_PROD is in <Ready> & DEV7 <= Vacuum 1 SP (TP006) & DEV3 closed feedback is available and CAT_DOSING_A

Step	Description	Action	Action Description
			in <Ready> and (CAT_DOSING_B is in Step <Ready> or <Standby>) then go to step S0200
S0200	Fill	A0010	Set Task Pointer
		A0020	Set Product Code (TP003) & Batch Number (TP004)
		A0030	Stopping DEV1
		A0040	Enabling Conditional Logic for DEV2
		A0050	Closing and Disabling Conditional Logic for DEV3
		A0060	Closing and Disabling Conditional Logic for DEV4
		A0070	Disabling Conditional Logic & Opening DEV5
		A0080	Closing and Disabling Conditional Logic for DEV6
		A0090	Set DEV8 Mode to Remote Out
		A0100	Set DEV8 output to Closed and Disable Conditional Logic
		A0110	Closing and Disabling Conditional Logic for DEV9
		A0120	Closing and Disabling Conditional Logic for DEV10
		A0130	Wait Message
T0300A	Heating		If DERTN_PROD is in <Drain Wait> then go to step S0300
S0300	Heating	A0010	Set Task Pointer
		A0020	Acquiring Reactor 3
		A0030	Starting Heat command
		A0040	Starting DEV1
		A0050	Closing and Disabling Conditional Logic for DEV2
		A0060	Closing and Disabling Conditional Logic for DEV3
		A0070	Closing and Disabling Conditional Logic for DEV4
		A0080	Disabling Conditional Logic & Opening DEV5
		A0090	Closing and Disabling Conditional Logic for DEV6
		A0100	Set DEV8 Mode to Remote Out
		A0110	Set DEV8 output to Closed and Disable Conditional Logic
		A0120	Closing and Disabling Conditional Logic for DEV9

Step	Description	Action	Action Description
		A0130	Closing and Disabling Conditional Logic for DEV10
T0400A	CatalystHYD		If DEV7 >= Setpoint - Minimum Temperature (TP007) then go to step S0400
S0400	CatalystHYD	A0010	Set Task Pointer
		A0020	Starting DEV1
		A0030	Closing and Disabling Conditional Logic for DEV2
		A0040	Closing and Disabling Conditional Logic for DEV3
		A0050	Closing and Disabling Conditional Logic for DEV4
		A0060	Closing and Disabling Conditional Logic for DEV5
		A0070	Opening and Disabling Conditional Logic for DEV6
		A0080	Set DEV8 Mode to Remote Out
		A0090	Set DEV8 output to Closed and Disable Conditional Logic
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Enabling Conditional Logic for DEV10
T0500A	Parameter Check		If CAT_DOSING_A is in <Drain Wait> or CAT_DOSING_B is in <Drain Wait> then go to step S0500
S0500	Parameter Check	A0010	Starting DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Set DEV8 Mode to Remote Out
		A0090	Set DEV8 output to Closed and Disable Conditional Logic
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Closing and Disabling Conditional Logic for DEV10
		A0120	Reset <Parameter Check> Button (OP003)
T0600A	ReactionHYD		If Operator select <ReactionHYD> command (OP002) & DEV7 < Vacuum 1 SP

Step	Description	Action	Action Description
			(TP006) & DEV2 and DEV3 closed feedback & DEV4, DEV5, DEV6 closed feedback is available then go to step S0600
S0600	ReactionHYD	A0010	Starting DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Set DEV7 Mode to Remote Cascade
		A0080	Setting DEV7 setpoint and Enabling Conditional Logic
		A0090	Enabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0120	Reset <Add H2> button (OP005)
T0700A	Reaction Fail		If DEV7 or FT_203 or PSH_203 or RD_2031 is in fault then go to step S0700
T0800B	SelectionHYD		If required H2 is added or Operator Select <Stop Reaction> command (OP004) then go to step S0800
S0700	Reaction Fail	A0010	Starting DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Set DEV8 Mode to Remote Out
		A0090	Set DEV8 output to Closed and Disable Conditional Logic
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Closing and Disabling Conditional Logic for DEV10
T0800A	SelectionHYD		If Operator Select <Stop Reaction> command (OP004) then go to step S0800

Step	Description	Action	Action Description
S0800	Selection HYD	A0010	Starting DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Set DEV8 Mode to Remote Out
		A0090	Set DEV8 output to Closed and Disable Conditional Logic
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Closing and Disabling Conditional Logic for DEV10
		A0130	Reset Button <Stop Reaction> (OP004)
T0500B	Parameter Check		If Operator Select <Parameter Check> command (OP003) then go to step S0500
T0600B	Reaction HYD		If Operator Select <Add H2> command and "Parameter Check OK" (DEV7 < HH & not PSH_203 triggered & not RD_2031 triggered & DEV2, DEV3, DEV4, DEV5, DEV6 closed FB) then go to step S0600
T9900	Complete		If Operator Select <Drain> command and DEV7 < 6 bar then go step S9900
S9900	Complete	A0010	Hydrogenation Command Complete and change Owner ID
END	Command Complete		True

Table 20: Hydrogenation Sequence

3.3.3

Command 02 – Drain

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Drain Command
T0100	Drain Check		If Task Pointer = 8 then go to step S0100
T0200B	Drain		If Task Pointer = 9 then go to step S0200
T0300B	Standby		If Task Pointer = 10 then go to step S0300
S0100	Drain Check	A0020	Starting DEV1
		A0030	Closing and Disabling Conditional Logic for DEV2

Step	Description	Action	Action Description
		A0040	Closing and Disabling Conditional Logic for DEV3
		A0050	Enabling Conditional Logic for DEV4
		A0060	Closing and Disabling Conditional Logic for DEV5
		A0070	Enabling Conditional Logic for DEV6
		A0080	Set DEV8 Mode to Remote Out
		A0090	Set DEV8 output to Closed and Disable Conditional Logic
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Closing and Disabling Conditional Logic for DEV10
		A0120	Set Message
T0200A	Drain		If TT_203 <= Drain Temperature Setpoint (TP008) & DEV7 <= Vacuum SP 2 (TP009) & REACTOR2 is in <Standby> & LS_2382B is in alarm + 10 secs delay then go to step S0200
S0200	Drain	A0010	Checking owner ID of Reactor2 and Set Task Pointer
		A0020	Acquiring Reactor2
		A0030	Start Command to Reactor2
		A0040	Stopping DEV1
		A0050	Closing and Disabling Conditional Logic for DEV2
		A0060	Enabling Conditional Logic for DEV3
		A0070	Closing and Disabling Conditional Logic for DEV4
		A0080	Closing and Disabling Conditional Logic for DEV5
		A0090	Opening and Disabling Conditional Logic for DEV6
		A0100	Set DEV8 Mode to Remote Out
		A0110	Set DEV8 output to Closed and Disable Conditional Logic
		A0120	Closing and Disabling Conditional Logic for DEV9
		A0130	Closing and Disabling Conditional Logic for DEV10
		A0140	Set Message
T0300A	Standby		If LSH_282B or LSL_203 is in alarm + 5 sec delay then go to step S0300
S0300	Standby	A0030	Stopping DEV1

Step	Description	Action	Action Description
		A0040	Closing and Disabling Conditional Logic for DEV2
		A0050	Closing and Disabling Conditional Logic for DEV3
		A0060	Closing and Disabling Conditional Logic for DEV4
		A0070	Enabling Conditional Logic for DEV5
		A0080	Closing and Disabling Conditional Logic for DEV6
		A0090	Set DEV8 Mode to Remote Out
		A0100	Set DEV8 output to Closed and Disable Conditional Logic
		A0110	Closing and Disabling Conditional Logic for DEV9
		A0120	Closing and Disabling Conditional Logic for DEV10
T0400	Disable HM		If no pending actions in S0300 then proceed step S0400
S0400	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0500	Release All Devices		If no pending actions in S0400 then proceed step S0500
S0500	Release All Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	Release Private Device
		A0100	Release Private Device
		A0110	Release Private Device
		A0120	Release Private Device
T9900	Complete		If no pending actions in S0500 then proceed step S9900
S9900	Complete	A0010	Set Message: Drain Command Complete
END	Command Complete		True

Table 21: Drain Sequence

3.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	RESET	Boolean	NA	False
OP002	Reaction Hyd	Boolean	NA	False
OP003	PARAM Check	Boolean	NA	False
OP004	Stop Reaction	Boolean	NA	False
OP005	Add H2	Boolean	NA	False
OP006	MODE	Floating Point	NA	MAN = 0 (AUTO = 1)
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Venting time	Floating Point	sec	20
TP003	Product Code	String	NA	NONE
TP004	Batch Number	Floating Point	NA	10
TP006	Vaccum Setpoint 1	External Reference	NA	HY1_RECIP/VA CUUM_SP_1_7
TP007	Tmin Setpoint	External Reference	NA	HY1_RECIP/MI N_TEMP_7
TP008	Drain Temp SP	External Reference	NA	HY1_RECIP/D RAIN_TEMP
TP009	Vaccum Setpoint 2	External Reference	NA	HY1_RECIP/VA CUUM_SP_2_7
TP010	Tmax Setpoint	Floating Point	°C	1.2

Table 22: Operating and Tuning Parameters

3.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		HY1EQ_1303
OPR_CMD_MASK		3
PT_203	External Reference	HY1_PT1203/PV
LS_2382B	External Reference	HY1_LS12382B/DI1/DISC_ACT
LSH_282B	External Reference	HY1_LSH1282B/DI1/DISC_ACT

Parameter Name	Parameter Type	Value/Reference
LSL_203	External Reference	HY1_LSL1203/DI1/DISC_ACT
ESM	String	ESM_HY_01
DETRN_PROD	String	HY1EQ_1303_82A
CAT_DOSING_A	String	HY1EQ_1355A
CAT_DOSING_B	String	HY1EQ_1355B
REACTOR2	String	HY1EQ_1382B
DY_REF_1	Dynamic Reference	
DY_REF_2	Dynamic Reference	
DY_REF_3	Dynamic Reference	
DY_REF_4	Dynamic Reference	
DY_REF_5	Dynamic Reference	
DY_REF_6	Dynamic Reference	
PSH_203	String	HY1_PSH1203
RD_2031	String	HY1_RD12031
FT_203	String	HY1_FT1203H
TT_203	External Reference	HY1_TT1203/PV
REACTOR3	String	HY1EQ_1303_HCC
HOLD	External Reference	HY1EQ_1300/OP003_VALUE
DRAIN	Boolean	
DELAY_TMR	Floating Point	10
DRN_1335B	External Reference	HY1EQ_1316A/OP002_VALUE

Table 23: Configurable Parameters

3.6

Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_HY_REACTOR1	Hydrogenation Area: Hydrogenation Reactor

Table 24: Named Set Details

Details of states for NS_HY_REACTOR1 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Hydrogenation
02	Drain

Table 25: Name Set State Detail

4 EQUIPMENT MODULE CLASS DESIGN – EM_HY_REACTOR3 (1-1303-HCC)

4.1 Functional Description

EM_HY_REACTOR3 is a class based equipment module. EM_HY_REACTOR3 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Heat Command and shall be released on completion of Heat Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

4.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_HY_REACTOR3	HY1EQ_1303_HCC
DEV1	_V12_IP_130_SE	HY1_V1203
DEV2	_V12_IP_130_SE	HY1_V1203S
DEV3	_V12_IP_130_SE	HY1_V1203WR
DEV4	_V12_IP_130_SE	HY1_V1303WV
DEV5	_V12_IP_130_SE	HY1_V1203E
DEV6	_CSD_130_SE	HY1_FT1078TW
DEV7	_CMT_130_SE	HY1_TT1203

Table 26: Instances

4.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

4.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out & Hold=True		Go to step H1000
H1000	Hold	A0010	Closing DEV1

Step	Description	Action	Action Description
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing DEV4
		A0050	Closing DEV5
		A0060	Set DEV6 Mode to Remote Out
		A0070	Set DEV6 output to Closed
T9900A	Actions Complete or Time Out		Go to step H9900A
T9900B	Actions Complete or Time Out		Go to step H9900B
H9900A	Hold Complete	A0010	Completed
END	Command Complete		True
H9900B	Release All Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	Reset Parameters
END1	Command Complete		True

Table 27: Hold Sequence

4.3.2 Command 01 – Heat

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Heat Command
T0010	Initializations		If First Pass = False then go to step S0010
T0020A	Acquire Devices		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020



Step	Description	Action	Action Description
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Acquire Private Device
		A0060	Acquire Private Device
		A0070	Acquire Private Device
		A0080	Acquire Private Device
		A0090	Reset HOLD_REQ
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0200B	Heat		If Task Pointer = 2 then go to step S0200
T0300B	Vent		If Task Pointer = 3 then go to step S0300
T0400C	Cool		If Task Pointer = 4 then go to step S0400
T0500B	Drain		If Task Pointer = 5 then go to step S0500
S0100	Standby	A0010	Set Task Pointer
		A0020	Closing DEV1
		A0030	Closing and Disabling Conditional Logic for DEV2
		A0040	Opening DEV3
		A0050	Closing DEV4
		A0060	Closing DEV5
		A0070	Set DEV6 Mode to Remote Out
		A0080	Set DEV6 output to Closed
T0200A	Heat		If DEV7 < Tmin Setpoint (TP004) & REACTOR1 is in step <Heating> then go to step S0200
T0400B	Cool		If (DEV7 > Tmax Setpoint (TP005) during REACTOR1 is in step <ReactionHYD> or <Reaction Fail> or <Selection HYD>) or (DEV7 > Drain Temperature Setpoint (TP006) during REACTOR1 in step <DrainCheck>) then go to step S0400
S0200	Heat	A0010	Opening DEV1 and Set Task Pointer
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing DEV4
		A0050	Closing DEV5
		A0060	Set DEV6 Mode to Remote Out
		A0070	Set DEV6 output to Closed
		A0080	Wait Message for Temperature
T0300A	Vent		If DEV7 >= Tmin Setpoint (TP004) + 2deg C hysteresis then go to step S0300

Step	Description	Action	Action Description
S0300	Vent	A0010	Opening DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Opening DEV4
		A0050	Closing DEV5
		A0060	Set DEV6 Mode to Remote Out
		A0070	Set DEV6 output to Closed
		A0080	Set Venting Time
		A0090	Start Venting Time
		A0100	Venting Time Message
T0400A	Cool		If Venting Time (TP002) Over then go to step S0400
S0400	Cool	A0010	Set Task Pointer
		A0020	Closing DEV1
		A0030	Closing and Disabling Conditional Logic for DEV2
		A0040	Opening DEV3
		A0050	Closing DEV4
		A0060	Closing DEV5
		A0070	Set DEV6 Mode to Remote Out
		A0080	Set DEV6 output to 50%
		A0090	Set DEV6 Mode to Manual
T0500A	Drain		If (DEV7 <= Max Temperature Setpoint (TP005) - 10 degC hysteresis during 1-1303 Step <ReactionHYD> or <ReactionFail> or <SelectionHYD>) or (DEV7 <= Drain Temperature Setpoint (TP006) during REACTOR1 Step <Drain>) then go to step S0500
S0500	Drain	A0010	Closing DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing DEV4
		A0050	Opening DEV5
		A0060	Set DEV6 Mode to Remote Out
		A0070	Set DEV6 output to Closed
		A0080	Set Drain Time
		A0090	Start Drain Time
		A0100	Drain Time Message

Step	Description	Action	Action Description
T0600	Close Devices		If Drain time (TP001) is over then go to step S0600
S0600	Close Devices	A0010	Closing DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing DEV4
		A0050	Closing DEV5
		A0060	Set DEV6 Mode to Remote Out
		A0070	Set DEV6 output to Closed
T0700	Disable HM		If no pending actions in S0600 then go to step S0700
S0700	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0800	Release All Devices		If no pending actions in S0700 then go to step S0800
S0800	Release All Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
T9900	Complete		If no pending actions in S0800 then go to step S9900
S9900	Complete	A0010	Set Message: Heat Command Complete
		A0020	Self Release the EM and First Pass=False

Table 28: Heat Sequence

4.4

Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	RESET	Boolean	NA	False
OP002	MODE	Floating Point	NA	MAN = 0 (AUTO = 1)

Parameter	Description	Type	Units	Default Value/Reference
Tuning Parameters				
TP001	Delay Time	Floating Point	sec	5
TP002	Venting time	Floating Point	sec	20
TP003	FCV_078TW RSP	String	Kg/h	0
TP004	TT_203 Min Temp	External Reference	°C	HY1_RECIP/MIN_TEMP_7
TP005	TT_203 Max Temp	External Reference	°C	HY1_RECIP/MAX_TEMP_7
TP006	Drain Temp SP	External Reference	°C	HY1_RECIP/DRAIN_TEMP_7

Table 29: Operating and Tuning Parameters

4.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		HY1EQ_1303_HCC
OPR_CMD_MASK		3
HOLD	External Reference	HY1EQ_1300/OP003_VALUE
DY_REF_1	Dynamic Reference	
DY_REF_2	Dynamic Reference	
DY_REF_3	Dynamic Reference	
DY_REF_4	Dynamic Reference	
DY_REF_5	Dynamic Reference	
REACTOR1	String	HY1EQ_1303
STEP_TIME	Floating Point	0
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False

Table 30: Configurable Parameters

4.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_HY_REACTOR3	Hydrogenation Area: Hydrogenation Reactor3

Table 31: Named Set Details

Details of states for NS_HY_REACTOR3 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Heat

Table 32: Name Set State Details

5 EQUIPMENT MODULE CLASS DESIGN – EM_HY_CAT_DOSING (1-1335A OR 1-1335B)

5.1 Functional Description

EM_HY_CAT_DOSING is a class based equipment module that drives three other sequences to complete Catalyst Dosing Process. EM_HY_CAT_DOSING is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold, Fill and Drain Command. Fill command shall be provided by EM_HY_DERTN_PROD during fill step. Drain command shall be provided by EM_HY_REACTOR1 during CatalystHYD. All devices used in EM shall be acquired in Fill Command and shall be released in Drain Command. Hold Request shall be enabled for all acquired devices at the end of HOLD command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

5.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name	
EM	EM_HY_CAT_DOSING	HY1EQ_1355A	HY1EQ_1355B
DEV1	_V12_IP_130_SE	HY1_V1235A	HY1_V1035BF
DEV2	_MRSS11_130_SE	HY1_AG1135A	HY1_AG1135B
DEV3	_V12_IP_130_SE	HY1_V1235AS	HY1_V1335BS
DEV4	_V12_IP_130_SE	HY1_V1135	HY1_V1235BD
DEV5	_MRSS11_130_SE	HY1_PU1135A	HY1_PU2135B
DEV6	_V12_IP_130_SE	HY1_V1235BN	HY1_V1235BBN
DEV7	_MRSS11_130_SE	HY1_PU1001	

Table 33: Instances

5.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

5.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Start	A0010	Store Fail Message
		A0020	Reset HOLD_REQ
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out		Go to step H1000

Step	Description	Action	Action Description
H1000	Hold Command 1	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional logic for DEV7
T9900A	Actions Complete or Time Out and Hold=True		Go to step H9900A
T9900B	Actions Complete or Time Out and Hold=False		Go to step H3000
H3000	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	Reset Parameters
END1	Command Complete		True
H9900	Command Hold End	A0010	Completed
END	Command Complete		True

Table 34: Hold Sequence

5.3.2 Command 01 – Fill

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Fill Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	Set Message: Place Holder
		A0030	First Pass = True
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Device	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Acquire Private Device
		A0060	Acquire Private Device
		A0070	Acquire Private Device
		A0080	Acquire Private Device
		A0110	Reset HOLD_REQ
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0200B	Fill		If Task Pointer = 2 then go to step S0200
S0100	Standby	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional Logic for DEV7
T0200A	Fill		If DERTN_PROD is in <Fill> then go to step S0200.
S0200	Fill	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3

Step	Description	Action	Action Description
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
T0300A	Filling Catalyst		If LSH_135 is in alarm then go to step S0300
S0300	Filling Catalyst	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional Logic for DEV7
T9900	Complete		If Operator select <Confirmed Catalyst Dosed> command (OP005) and TT_135 >= Setpoint Heating Temperature (OP004) then go to step S9900
S9900	Complete	A0010	Set Message: Fill Command Complete
END	Command Complete		True

Table 35: Fill

5.3.3 Command 02 – Drain

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Drain Command
T0100A	Ready		If no pending actions in S0000 then go to step S0100
T0100B	Ready		If Task Pointer = 1 then go to step S0100
T0200B	Drain		If Task Pointer = 2 then go to step S0200
T0300B	Blow		If Task Pointer = 3 then go to step S0300
T0400B	Drain Wait		If Task Pointer = 4 then go to step S0400
S0100	Ready	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3

Step	Description	Action	Action Description
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional Logic for DEV7
T0200A	Drain		If REACTOR1 is in <CatalystHYD> and 1335A is executed then 1335B should not be in <Ready>, <Drain>, & <Blow> (Vice versa) then go to step S0200
S0200	Drain	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Stoping and Disabling Conditional Logic for DEV7
		A0080	Set Drain Time (TP003)
		A0090	Start Drain Time
		A0100	Drain Time Message
T0300A	Blow		If LSL_135 in alarm with 10 Secs Delay time (TP002) and check Drain time (TP003) is completed only if 1-1335B is executing then go to step S0300
S0300	Blow	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Stoping and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Stoping and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Stoping and Disabling Conditional Logic for DEV7
		A0080	Set Blow Time (OP003)
		A0090	Start Blow Time
		A0100	Blow Time Message

Step	Description	Action	Action Description
T0400A	Drain Wait		If Blow time (OP003) is completed then go to step S0400
S0400	Drain Wait	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Stoping and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Stoping and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Stoping and Disabling Conditional Logic for DEV7
T0500	Disable HM & SM		If REACTOR1 is in <ParameterCheck> then go to step S0500
S0500	Disable HM & SM	A0010	Disable HM 01 (Module Ownership Lost)
T0600	Release Devices		If no pending actions in S0500 then go to step S0600
S0600	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	self release
T9900	Complete		If no pending actions in S0600 then go to step S9900
S9900	Complete	A0010	Set Message: Drain Command Complete
END	Command Complete		True

Table 36: Drain Sequence

5.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value / Reference
Operating Parameters				
OP001	MODE	Boolean	NA	MAN = 0 (AUTO = 1)
OP002	RESET	Boolean	NA	False
OP003	Blow Time	Floating Point	sec	0
OP004	Heating Temperature Setpoint TT_135	Floating Point	°C	0
OP005	Confirm Catalyst Dosed	Named Set	NA	YES
OP006	EM_HY_REACTO R1	String	NA	0
Tuning Parameters				
TP001	Delay Timer 1	Floating Point	sec	5
TP002	Delay Timer 2	Floating Point	sec	10
TP003	Drain Time	Floating Point	sec	5

Table 37: Operating and Tuning Parameters

5.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference	
MY_TAG		HY1EQ_1355A	HY1EQ_1355B
OPR_CMD_MASK		7	7
ESM	String	ESM_HY_01	
LSH_135	External Reference	HY1_LSH1135A/DI1/DISC_ACT	HY1_LSH1135B/DI1/DISC_ACT
LSL_135	External Reference	HY1_LSL1135A/DI1/DISC_ACT	HY1_LSL1135B/DI1/DISC_ACT
TT_135	External Reference	HY1_TT1135A/PV	HY1_TT1135B/PV
V_235	External Reference	HY1_V1235/PV_D	
DERTN_PROD	String	HY1EQ_1303_82A	

Parameter Name	Parameter Type	Value/Reference
REACTOR1	String	HY1EQ_1303
DY_REF_1	Dynamic Reference	
STEP_TIME	Floating Point	
COMPLETED_TIME	Floating Point	
REET_COMP_TM	Boolean	
HOLD	External Reference	HY1EQ_1300/OP003_VALUE
COMMAND_1335	External Reference	HY1EQ_1302_82A/OP003_VALUE

Table 38: Configurable Parameters

5.6

Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_HY_CAT_DOSING	Hydrogenation Area: Catalyst Dosing EM

Table 39: Named Set Details

Details of states for NS_HY_CAT_DOSING named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Fill
02	Drain

Table 40: Name Set State Details

6 EQUIPMENT MODULE CLASS DESIGN – EM_HY_REACTOR2 (1-1382B)

6.1 Functional Description

EM_HY_REACTOR2 is a class based equipment module that communicates with five other sequences to complete hydrogenation process in a reactor. EM_HY_REACTOR2 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. HY1EQ-1303 (EM) shall provide start command during drain step. Initially Operator shall be provided only with Hold and Start Command. All devices used in EM shall be acquired in initial steps of Start Command and shall be released in last step of Start Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

6.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_HY_REACTOR2	HY1EQ_1382B
DEV1	_MRSS11_130_SE	HY1_AG1182B
DEV2	_MRSS11_130_SE	HY1_PU1282B
DEV3	_V12_IP_130_SE	HY1_V1381BD
DEV4	_V12_IP_130_SE	HY1_V1282BF1
DEV5	_V12_IP_130_SE	HY1_V1382CF3
DEV6	_V12_IP_130_SE	HY1_V1182BBN
DEV7	_V12_IP_130_SE	HY1_V1381AF
DEV8	_V12_IP_130_SE	HY1_V1381ABN
DEV9	_CSD_130_SE	HY1_TT1281A
DEV10	_CSD_130_SE	HY1_FT1282B
DEV11	EM_HY_FILTER	HY1EQ_1316A
DEV12	EM_HY_REACT_FILT	HY1EQ_1382C
DEV13	_V12_IP_130_SE	HY1_V1203D
DEV14	EM_HY_FILTER_B	HY1EQ_1316B

Table 41: Instances

6.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

6.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Common Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Set DEV9 Mode to Remote Out
		A0100	Set DEV9 output to Close and Disable Conditional Logic
		A0110	Set DEV10 Mode to Remote Out
		A0120	Set DEV10 output to Close and Disable Conditional Logic
		A0130	Closing DEV13
T9900A	Actions Complete or Time Out & Hold=True		Go to step H9900A

Step	Description	Action	Action Description
T9900B	Actions Complete or Time Out & Hold=False		Go to step H9900B
H9900A	Common Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	Release Private Device
		A0100	Release Private Device
		A0110	Release Private Device
		A0120	Release Private Device
		A0130	Reset Parameters
END1	Command Complete		True

Table 42: Hold Sequence

6.3.2 Command 01 – Start

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Start Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020

Step	Description	Action	Action Description
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Acquire Private Device
		A0060	Acquire Private Device
		A0070	Acquire Private Device
		A0080	Acquire Private Device
		A0090	Acquire Private Device
		A0100	Acquire Private Device
		A0110	Acquire Private Device
		A0120	Acquire Private Device
		A0130	Acquire Private Device
		A0140	Reset HOLD_REQ
T0100	Standby		Go to step S0100
T0200B	Fill		If Task Pointer = 2 then go to step S0200
T0300B	Ready		If Task Pointer = 3 then go to step S0300
T0400B	Drain		If Task Pointer = 4 then go to step S0400
T0500B	Blow 1		If Task Pointer = 5 then go to step S0500
T0600B	Blow 2		If Task Pointer = 6 then go to step S0600
T0700B	Drain Wait		If Task Pointer = 7 then go to step S0700
S0100	Standby	A0010	Set Message: Reset product code & batch number
		A0020	Closing and Disabling Conditional Logic for DEV1
		A0030	Closing and Disabling Conditional Logic for DEV2
		A0040	Closing DEV3
		A0050	Closing and Disabling Conditional Logic for DEV4
		A0060	Closing and Disabling Conditional Logic for DEV5
		A0070	Closing and Disabling Conditional Logic for DEV6
		A0080	Closing and Disabling Conditional Logic for DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8

Step	Description	Action	Action Description
		A0100	Set DEV9 Mode to Remote Out
		A0110	Set DEV9 output to Closed and Disable Conditional Logic
		A0120	Set DEV10 Mode to Remote Out
		A0130	Set DEV10 output to Closed and Disable Conditional Logic
		A0140	Closing DEV13
T0200A	Fill		If REACTOR1 is in step <Drain> then go to step S0200
S0200	Fill	A0010	Set Message: Copy Product Code & Batch Number and Set Task Pointer
		A0020	Enabling Conditional Logic for DEV1
		A0030	Closing and Disabling Conditional Logic for DEV2
		A0040	Closing DEV3
		A0050	Closing and Disabling Conditional Logic for DEV4
		A0060	Closing and Disabling Conditional Logic for DEV5
		A0070	Closing and Disabling Conditional Logic for DEV6
		A0080	Closing and Disabling Conditional Logic for DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Set DEV9 Mode to Remote Out
		A0110	Set DEV9 output to Closed and Disable Conditional Logic
		A0120	Set DEV10 Mode to Remote Out
		A0130	Set DEV10 output to Closed and Disable Conditional Logic
		A0140	Opening DEV13
		A0150	Set Message: Reactor1 is not in drain
T0300A	Ready		If REACTOR1 is not in step Drain and (LSL_203 is in alarm or LSH_282B in alarm) with 5 sec delay then go to step S0300
S0300	Ready	A0005	Set Task Pointer
		A0010	Check DEV11 is not owned
		A0020	Start filtration command of DEV11

Step	Description	Action	Action Description
		A0030	Check DEV12 is not owned
		A0040	Start Process Filter command of DEV12
		A0050	Enabling Conditional Logic for DEV1
		A0060	Closing and Disabling Conditional Logic for DEV2
		A0070	Closing DEV3
		A0080	Closing and Disabling Conditional Logic for DEV4
		A0090	Closing and Disabling Conditional Logic for DEV5
		A0100	Closing and Disabling Conditional Logic for DEV6
		A0110	Closing and Disabling Conditional Logic for DEV7
		A0120	Closing and Disabling Conditional Logic for DEV8
		A0130	Set DEV9 Mode to Remote Out
		A0140	Set DEV9 output to Closed and Disable Conditional Logic
		A0150	Set DEV10 Mode to Remote Out
		A0160	Set DEV10 output to Closed and Disable Conditional Logic
		A0170	Close DEV13
		A0180	Set Message: Ready to Drain and Dynamic Reference
		A0190	Check DEV14 is not owned
		A0200	Start Process Filter command of DEV14
T0400A	Drain		If My Product == REACT_FILT Product and REACT_FILT in steps Fill and LSH_282C1 not in alarm and ((DERTN_PROD_FILL is in step Fill or (DERTN_PROD_FILL Standby and Startup step in Drain)) then go to step S0400
S0400	Drain	A0010	Set Message: Copy Product Code & Batch Number and Set Task Pointer
		A0020	Enabling Conditional Logic for DEV1
		A0030	Enabling Conditional Logic for DEV2
		A0040	Opening DEV3
		A0050	Enabling Conditional Logic for DEV4
		A0060	Enabling Conditional Logic for DEV5

Step	Description	Action	Action Description
		A0070	Closing and Disabling Conditional Logic for DEV6
		A0080	Enabling Conditional Logic for DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Set DEV9 Mode to Remote Cascade
		A0110	Setting DEV9 setpoint to (TT_281A Setpoint) and Enabling Conditional Logic
		A0120	Set DEV10 Mode to Remote Cascade
		A0130	Setting DEV10 setpoint to (FT_282B Setpoint) and Enabling Conditional Logic
		A0140	Closing DEV13
		A0150	Check for LSL_282B is active
T0500A	Blow1		If LSL_2382B in alarm + 5 sec delay then go to step S0500
S0500	Blow1	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Opening and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Set DEV9 Mode to Remote Out
		A0100	Set DEV9 output to Close and Disable Conditional Logic
		A0110	Set DEV10 Mode to Remote Out
		A0120	Set DEV10 output to Open and Disable Conditional Logic
		A0130	Closing DEV13
		A0140	Set Blow 1Time
		A0150	Start Blow 1 Time
		A0160	Blow 1 Time Message

Step	Description	Action	Action Description
T0600A	Blow2		If Blow 1 Time is over then go to step S0600
S0600	Blow2	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Set DEV9 Mode to Remote Out
		A0100	Set DEV9 output to Close and Disable Conditional Logic
		A0110	Set DEV10 Mode to Remote Out
		A0120	Set DEV10 output to Open and Disable Conditional Logic
		A0130	Closing DEV13
		A0140	Set Blow 2 Time
		A0150	Start Blow 2 Time
		A0160	Blow 2 Time Message
T0700A	Drain Wait		If Blow 2 Time is over then go to step S0600
S0700	Drain Wait	A0010	Saving Step time and Set Task Pointer
		A0020	Stopping and Disabling Conditional Logic for DEV1
		A0030	Closing and Disabling Conditional Logic for DEV2
		A0040	Closing DEV3
		A0050	Closing and Disabling Conditional Logic for DEV4
		A0060	Closing and Disabling Conditional Logic for DEV5
		A0070	Closing and Disabling Conditional Logic for DEV6

Step	Description	Action	Action Description
		A0080	Closing and Disabling Conditional Logic for DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Set DEV9 Mode to Remote Out
		A0110	Set DEV9 output to Close and Disable Conditional Logic
		A0120	Set DEV10 Mode to Remote Out
		A0130	Set DEV10 output to Close and Disable Conditional Logic
		A0140	Closing DEV13
		A0150	Waiting for Delay Timer
T0800	Disable HM		If no pending actions in step S0700 then go to step S0800 after 5 sec delay
S0800	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0900	Release Devices		If no pending actions in S0800 then go to step S0900
S0900	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	Release Private Device
		A0100	Release Private Device
		A0110	Release Private Device
		A0120	Release Private Device
T9900	Complete		If no pending actions in S0900 then go to step S9900
S9900	Complete	A0010	Set Message: Start Command Complete
		A0020	Self Release the EM
END	Command Complete		True

Table 43: Start Sequence

6.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	RESET	Boolean		False
OP002	MODE	Floating Point		AUTO = 1 MAN = 0
Tuning Parameters				
TP001	Product Code	String	NA	0
TP002	Batch Number	Floating Point	NA	0
TP003	Blow 1 Time	Floating Point	sec	0
TP004	Blow 2 Time	Floating Point	sec	0

Table 44: Operating and Tuning Parameters

6.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		HY1EQ_1382B
OPR_CMD_MASK		3
HOLD	External Reference	HY1EQ_1300/OP003_VALUE
LSL_203	External Reference	HY1_LSL1203/DI1/DISC_ACT
LSL_2382B	External Reference	HY1_LS12382B/DI1/DISC_ACT
LSH_282C1	External Reference	HY1_LSH1282C1/DI1/DISC_ACT
LSH_282B	External Reference	HY1_LSH1282B/DI1/DISC_ACT
DERTN_PROD_FILL	String	HY1EQ_1303_82A
REACTOR1	String	HY1EQ_1303
REACT_FILT	String	HY1EQ_1382C
ESM	String	ESM_HY_01
DY_REF_1	Dynamic Reference	
DY_REF_2	Dynamic Reference	
DY_REF_3	Dynamic Reference	
DY_REF_4	Dynamic Reference	
DY_REF_5	Dynamic Reference	
DY_REF_6	Dynamic Reference	
DY_REF_7	Dynamic Reference	
STARTUP_DRAIN	String	HY1EQ_1300

Parameter Name	Parameter Type	Value/Reference
DELAY_TMR	Floating point	
STEP_TIME	Floating point	0
COMPLETED_TIME	Floating point	0
RESET_COMP_TM	Boolean	False

Table 45: Configurable Parameters

6.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_HY_REACTOR2	Hydrogenation Area: Hydrogenation Reactor2

Table 46: Named Set Details

Details of states for NS_HY_REACTOR2 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Start

Table 47: Name Set State Details

7 EQUIPMENT MODULE CLASS DESIGN – EM_HY_REACTOR4 (1-1382C)

7.1 Functional Description

EM_HY_REACTOR4 is a class based equipment module that executes hydrogenation reactor operations. EM_HY_REACTOR4 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Process Filter Command and shall be released in Blow Command. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as Graphics Buttons wherever possible.

7.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_HY_REACTOR4	HY1EQ_1382C
DEV1	_MRSS11_130_SE	HY1_AG1282C
DEV2	_MRSS11_130_SE	HY1_PU1282C
DEV3	_V12_IP_130_SE	HY1_V1382CD
DEV4	_V12_IP_130_SE	HY1_V1382CBN
DEV5	_V12_IP_130_SE	HY1_1316ABN
DEV6	_V12_IP_130_SE	HY1_V1116AY

Table 48: Instances

7.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

7.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Common Hold Start	A0010	Store Fail Message
		A0020	Reset HOLD_REQ
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Stopping and Disabling Conditional Logic for DEV1

Step	Description	Action	Action Description
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
T9900A	Actions Complete or Time Out & Hold=True		Go to step H9900A
T9900B	Actions Complete or Time Out & Hold=False		Go to step H9900B
H9900A	Common Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Disable HM 01 (Module Ownership Lost)
		A0020	Release Devices Message
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Reset Parameters
END1	Command Complete		True

Table 49: Hold Sequence

7.3.2

Command 01 – Process Filter

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Process Reactor Command
T0010	First Pass = False		If First Pass = False then go to step S0010

Step	Description	Action	Action Description
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Acquire Private Device
		A0060	Acquire Private Device
		A0070	Acquire Private Device
		A0080	Reset Hold Request
T0100	Standby		Go to step S0100
T0200B	Fill		If Task Pointer = 2 then go to step S0200
T0300B	Filter Aid		If Task Pointer = 3 then go to step S0300
T0400B	Filtration		If Task Pointer = 4 then go to step S0400
T0500B	Recirculation Filter		If Task Pointer = 5 then go to step S0500
T0600B	Drain Filter		If Task Pointer = 6 then go to step S0600
T0700B	Empty1		If Task Pointer = 7 then go to step S0700
T0800B	DrainWait 1		If Task Pointer = 8 then go to step S0800
T0900C	Empty2		If Task Pointer = 9 then go to step S0900
T1000C	Blow1		If Task Pointer = 10 then go to step S1000
T1100B	Blow2		If Task Pointer = 11 then go to step S1100
T1200B	DrainWait 2		If Task Pointer = 12 then go to step S1200
T1300B	Close		If Task Pointer = 13 then go to step S1300
S0100	Standby	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5



Step	Description	Action	Action Description
		A0060	Closing DEV6
		A0070	Reset product code & batch number
		A0080	Set Message
T0200A	Fill		If REACTOR2 is in step Ready then go to step S0200
S0200	Fill	A0020	Copy Product Code & Batch Number
		A0030	Enabling Conditional Logic for DEV1
		A0040	Stopping and Disabling Conditional Logic for DEV2
		A0050	Closing DEV3
		A0060	Closing and Disabling Conditional Logic for DEV4
		A0070	Closing and Disabling Conditional Logic for DEV5
		A0080	Closing DEV6
T0300A	Filter Aid		If LT_282C1 > Filtration Level (TP004) then go to step S0300
S0300	Filter Aid	A0010	Prompt: Start Filter Aid Process
		A0020	Enabling Conditional Logic for DEV1
		A0030	Stopping and Disabling Conditional Logic for DEV2
		A0040	Closing DEV3
		A0050	Closing and Disabling Conditional Logic for DEV4
		A0060	Closing and Disabling Conditional Logic for DEV5
		A0070	Closing DEV6
T0400A	Filtration		If (Operator Selection <Filter Aid ON> and if selected ENABLE FILTER AID ON, Operator Command <ACK Filter Aid>) or (Operator Selection <Filter Aid OFF>) then go to step S0400
S0400	Filtration	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4

Step	Description	Action	Action Description
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
T0200C	Fill		If ((Product REACTOR2 == Product & REACTOR2 not in step Standby) or (Product REACTOR1 == Product & REACTOR2 in Step Standby)) and FILTER is in step Drain then go to step S0200
T0500A	Recirculate Filter		If FILTER is in step Final Recirculation then go to step S0500
S0500	Recirculate Filter	A0010	Enabling Conditional Logic for DEV1
		A0020	Disabling Conditional Logic for DEV2
		A0025	Enabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Set Message:Filter is in Drain step
T0600A	Drain Filter		If FILTER is in step N2 Filtration then go to step S0600
S0600	Drain Filter	A0010	Enabling Conditional Logic for DEV1
		A0020	Stoping and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Set Message: Filter is not in Security Drain
T0700A	Empty 1		If FILTER is in step after Security Drain and HEEL_FILTRATN ON then go to step S0700.
T0900B	Empty 2		If FILTER is in step Security Drain and HEEL_FILTRATN OFF then go to step S0900

Step	Description	Action	Action Description
S0700	Empty 1	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Opening DEV6
		A0070	5 sec Delay
T0800A	DrainWait 1		If LSL_282C in alarm or LT_282C1 < LL setpoint (TP009) + 5 sec delay then go to step S0800
S0800	DrainWait 1	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Opening DEV6
T0900B	Empty 2		If FILTER_B is in step after Discharge step then go to step S0900
S0900	Empty 2	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Opening DEV6
		A0070	5 sec Delay
T1000A	Blow 1		If (LSL_282C in alarm or LT_282C1 < LL setpoint + 5 sec) and (LSL_316B in alarm when <Hell Filtration> OFF) then go to step S1000

Step	Description	Action	Action Description
S1000	Blow 1	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Opening and Disabling Conditional Logic for DEV6
		A0070	Set Blow1 Time
		A0080	Start Blow1 Time
		A0090	Blow1 Time Message
T1100	Blow 2		If Blow 1 Time is over then go to step S1100
S1100	Blow 2	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Set Blow2 Time
		A0080	Start Blow2 Time
		A0090	Blow2 Time Message
T1200	DrainWait 2		If Blow 2 Time is over then go to step S1200
S1200	DrainWait 2	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Opening DEV6

Step	Description	Action	Action Description
T1300A	Aux Off		If FILTER_B is in step Standby and FILTER is in step Standby Off//On then go to step S1300
S1300	Aux Off	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
T1400	Release Devices		If no pending actions in S1300 then go to step S1400
S1400	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
T9900	Complete		If no pending actions in S1400 then go to step S9900
S9900	Complete	A0010	Set Message: Process Filter Command Complete
		A0020	Self Release the EM & Reset First Pass
END	Command Complete		True

Table 50: Process Filter Sequence

7.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	RESET	Boolean		5
OP002	FILTER AID ON	Boolean		0
OP003	FILTER AID OFF	Boolean		0
OP004	MODE	Floating Point		AUTO = 1 MAN = 0
OP005	ACK Filter Aid	Boolean		0
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Blow 1 time	Floating Point	sec	0
TP003	Blow 2 time	Floating Point	sec	0
TP004	Filtration level	Floating Point	sec	0
TP005	Safe drain level	Floating Point	sec	0
TP006	Product Code	String	NA	
TP007	Batch Number	Floating Point	NA	0
TP008	Level to start Agitator	Floating Point	%	15
TP009	L Set Point	Floating Point	%	0

Table 51: Operating and Tuning Parameters

7.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		HY1EQ_1382C
OPR_CMD_MASK		3
ESM	String	ESM_HY_01/EM_DRN_FWD_TM
REACTOR2	String	HY1EQ_1382B
FILTER	String	HY1EQ_1316A
LT_282C1	External Reference	HY1_LT122382/A11/PV
LSL_282C	External Reference	HY1_LSL1282C/D11/DISC_ACT
DY_REF_1	Dynamic Reference	
DY_REF_2	Dynamic Reference	
DY_REF_3	Dynamic Reference	

Parameter Name	Parameter Type	Value/Reference
DY_REF_4	Dynamic Reference	
DY_REF_5	Dynamic Reference	
DY_REF_6	Dynamic Reference	
DY_REF_7	Dynamic Reference	
FLAG	Boolean	
HEEL_FILTRATN	External Reference	HY1EQ_1316B/OP002_VALUE
REACTOR1	String	HY1EQ_1303
STEP_TIME	Floating Point	0
HOLD	External Reference	HY1EQ_1300/OP003_VALUE
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False
LSL_316B	External Reference	HY1_LSL1316B/DI1/DISC_ACT
FILTER_B	String	HY1EQ_1316B

Table 52: Configurable Parameters

7.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_HY_REACTOR4	Hydrogenation Area: Reactor EM

Table 53: Named Set Details

Details of states for NS_HY_REACTOR4 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Process Reactor

Table 54: Name Set State Details

8 EQUIPMENT MODULE CLASS DESIGN – EM_HY_FILTER_A (1-1316A)

8.1 Functional Description

EM_HY_FILTER_A is a class based equipment module which communicates with other four sequences to complete filtration process of hydrogenation. EM_HY_FILTER_A is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Filtration commands. Filtration command shall be provided by EM_HY_REACTOR4 if EM_HY_FILTER_A is not owned by operator. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

8.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_HY_FILTER_A	HY1EQ_1316A
DEV1	_MRSS11_130_SE	HY1_PU1316A
DEV2	_V12_IP_130_SE	HY1_V1382CR
DEV3	_V12_IP_130_SE	HY1_V1316A2
DEV4	_V12_IP_130_SE	HY1_V1316AV3A
DEV5	_V12_IP_130_SE	HY1_V1316AV3B
DEV6	_V12_IP_130_SE	HY1_V1316AV3C
DEV7	_V12_IP_130_SE	HY1_V1316A3
DEV8	_V12_IP_130_SE	HY1_V1316A4
DEV9	_V12_IP_130_SE	HY1_V1316A6
DEV10	_V12_IP_130_SE	HY1_V1316A9
DEV11	_V12_IP_130_SE	HY1_V1316A10
DEV12	_V12_IP_130_SE	HY1_1316A1
DEV13	_V12_IP_130_SE	HY1_V1357CK

Table 55: Instances

8.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

Note : Mode Selection Philosophy,

MAN : Sequence Owned by Operator.

AUTO : Sequence Driven by Another Sequence.

8.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
T9900	Actions Complete or Time Out		Go to step H9900

Step	Description	Action	Action Description
H9900	Common Hold End	A0010	Completed

Table 56: Hold Sequence

8.3.2 Command 01 – Filtration

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Filtration Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Reset Parameter
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Device	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Acquire Private Device
		A0060	Acquire Private Device
		A0070	Acquire Private Device
		A0080	Acquire Private Device
		A0090	Acquire Private Device
		A0100	Acquire Private Device
		A0110	Acquire Private Device
		A0120	Acquire Private Device
		A0130	Acquire Private Device
		A0140	Acquire Private Device
		A0150	Reset Hold Request
T0100	Standby Off		Go to step S0100
T0200B	Standby On		If Task Pointer = 2 then go to step S0200
T0300B	Filling		If Task Pointer = 3 then go to step S0300
T0400B	Heating		If Task Pointer = 4 then go to step S0400
T0500B	Cleaning1		If Task Pointer = 5 then go to step S0500
T0600B	Cleaning2		If Task Pointer = 6 then go to step S0600
T0700B	Cleaning3		If Task Pointer = 7 then go to step S0700
T0800B	Re-filling		If Task Pointer = 8 then go to step S0800



Step	Description	Action	Action Description
T0900C	Precoat		If Task Pointer = 9 then go to step S0900
T1000B	Filtration		If Task Pointer = 10 then go to step S1000
T1100B	Command		If Task Pointer = 11 then go to step S1100
T1200B	Final Recirculation		If Task Pointer = 12 then go to step S1200
T1300B	Blow 1		If Task Pointer = 13 then go to step S1300
T1400B	Final Filtration		If Task Pointer = 14 then go to step S1400
T1500B	Blow 2		If Task Pointer = 15 then go to step S1500
T1600B	N2 Filtration		If Task Pointer = 16 then go to step S1600
T1700B	Blow 3		If Task Pointer = 17 then go to step S1700
T1800C	Security Drain		If Task Pointer = 18 then go to step S1800
T1900B	Depressurisation		If Task Pointer = 19 then go to step S1900
T2000B	Spent Route Check		If Task Pointer = 20 then go to step S2000
T2100B	Open V9		If Task Pointer = 21 then go to step S2100
T2200B	Discharge 1		If Task Pointer = 22 then go to step S2200
T2300B	Discharge 2		If Task Pointer = 23 then go to step S2300
T2400B	Discharge 3		If Task Pointer = 24 then go to step S2400
T2500B	Standby Open		If Task Pointer = 25 then go to step S2500
T2600B	Close V9		If Task Pointer = 26 then go to step S2600
S0100	Standby Off	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11

Step	Description	Action	Action Description
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
T0200A	Standby On		If (REACT_FILT is not in step <Standby> and mode is <Auto>) or (mode is <Manual> and Command <Standby ON>) then go to step S0200
S0200	Standby On	A0010	Set Task Pointer
		A0020	Stopping and Disabling Conditional Logic for DEV1
		A0030	Closing and Disabling Conditional Logic for DEV2
		A0040	Opening DEV3
		A0050	Closing and Disabling Conditional Logic for DEV4
		A0060	Closing and Disabling Conditional Logic for DEV5
		A0070	Closing and Disabling Conditional Logic for DEV6
		A0080	Closing and Disabling Conditional Logic for DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Closing and Disabling Conditional Logic for DEV10
		A0120	Closing and Disabling Conditional Logic for DEV11
		A0130	Closing and Disabling Conditional Logic for DEV12
		A0140	Closing and Disabling Conditional Logic for DEV13
T0300A	Filling		If (REACT_FILT is in step Filtration and DEV10 is closed and mode is <Auto>) or (mode is <Manual> and Command <Filling>) then go to step S0300
S0300	Filling	A0010	Set Task Pointer
		A0020	Copy Product Code and Increment Batches by 1
		A0030	Stopping and Disabling Conditional Logic for DEV1
		A0040	Closing and Disabling Conditional Logic for DEV2
		A0050	Opening DEV3

Step	Description	Action	Action Description
		A0060	Closing and Disabling Conditional Logic for DEV4
		A0070	Closing and Disabling Conditional Logic for DEV5
		A0080	Closing and Disabling Conditional Logic for DEV6
		A0090	Closing and Disabling Conditional Logic for DEV7
		A0100	Closing and Disabling Conditional Logic for DEV8
		A0110	Closing and Disabling Conditional Logic for DEV9
		A0120	Closing and Disabling Conditional Logic for DEV10
		A0130	Closing and Disabling Conditional Logic for DEV11
		A0140	Enabling Conditional Logic for DEV12
		A0150	Closing and Disabling Conditional Logic for DEV13
		A0160	Waiting for level LSH_316A in alarm
T0400A	Heating		If (LSH_316A is in alarm + 5 sec and mode is <Auto>) or (LSH_316A in alarm and mode is <Manual> and Command <Heating>) then go to step S0400
S0400	Heating	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Opening and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Opening and Disabling Conditional Logic for DEV6
		A0070	Opening and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13

Step	Description	Action	Action Description
		A0140	Set Heating Time
		A0150	Start Heating Time
		A0160	Blow 2 Time Message
T0500A	Cleaning1		If (Heating Time is over and mode is <Auto>) or (mode is <Manual> and Command <Cleaning1>) then go to step S0500
S0500	Cleaning1	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Opening and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Set Cleaning 1 Time
		A0150	Start Cleaning 1 Time
		A0160	Cleaning 1 Time Message
T0600A	Cleaning2		If (Cleaning Time is over and mode is <Auto>) or (mode is <Manual> and Command <Cleaning2>) then go to step S0600
S0600	Cleaning2	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5

Step	Description	Action	Action Description
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Opening and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Set Cleaning 2 Time
		A0150	Start Cleaning 2 Time
		A0160	Cleaning 2 Time Message
T0700A	Cleaning3		If (Cleaning Time is over and mode is <Auto>) or (mode is <Manual> and Command <Cleaning3>) then go to step S0700
S0700	Cleaning3	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Opening and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13

Step	Description	Action	Action Description
		A0140	Set Cleaning 3 Time
		A0150	Start Cleaning 3 Time
		A0160	Cleaning 3 Time Message
T0800A	Re-filling		If (Cleaning Time is over and mode is <Auto>) or (mode is <Manual> and Command <Re-filling>) then go to step S0800
S0800	Re-filling	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Waiting for level LSH_316A in alarm
T0900A	Precoat		If (LSH_316A in alarm + 5 secs and mode is <Auto>) or (LSH_316A in alarm and mode is <Manual> and Command <Precoat>) then go to step S0900
S0900	Precoat	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Opening and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5

Step	Description	Action	Action Description
		A0060	Opening and Disabling Conditional Logic for DEV6
		A0070	Opening and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Set Precoat Time
		A0150	Start Precoat Time
		A0160	Precoat Time Message
T1000A	Filtration		If (Precoat Tme is over and mode is <Auto>) or (mode is <Manual> and Command <Filtration>) then go to step S1000
S1000	Filtration	A0010	Enabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Opening and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Opening and Disabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
T0900B	Precoat		If mode is <Manual> and Command <Precoat> then go to step S0900
T1100A	Drain		If (mode is <Manual> and Command <Drain>) or (mode is <Auto> + (Max Batches

Step	Description	Action	Action Description
			(TP006) reached in filter or Max Filtration Pressure (TP007) reached)) then go to step S1100
T1200A	Final Recirculation		If (Product REACTOR2 not same as Product REACT_FILT (If REACTOR2 not in Standby) or Product REACTOR1 not same as Product REACT_FILT (If REACTOR2 in Standby) or (STARTUP Drain step and REACTOR1 in Standby and REACTOR2 in Standby)) and LT_2382 < LT_2382 Recycle SP (TP017_VALUE) and mode is <Auto> or mode is <Manual> and Command <Final Recirculation> then go to step S1200
S1100	Drain	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Opening and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Opening and Disabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Enabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Latch Drain Bit
T1300A	Blow1		If (LSL_316A in alarm + 5 sec delay and mode is <Auto>) or (mode is <Manual> and Command <Blow1>) then go to step S1300
S1200	Final Recirculation	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Opening and Disabling Conditional Logic for DEV4

Step	Description	Action	Action Description
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Opening and Disabling Conditional Logic for DEV6
		A0070	Opening and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Set Final Recirculation Time
		A0150	Start Final Recirculation Time
		A0160	Final Recirculation Time Message
T1400A	Final Filtration		If (Final Filtration Time is over and mode is <Auto>) or (mode is <Manual> and Command <Final Filtration>) then go to step S1400
S1300	Blow1	A0005	Unlatch Drain Bit
		A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Opening and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Disabling Conditional Logic for DEV9
		A0095	Enabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12

Step	Description	Action	Action Description
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Set Blow1 Time
		A0150	Start Blow1 Time
		A0160	Blow1 Time Message
		A0170	Unlatch Drain Bit
T1500	Blow2		If (Blow Time is over and mode is <Auto>) or (mode is <Manual> and Command <Blow2>) then go to step S1500
S1400	Final Filtration	A0010	Enabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Opening and Disabling Conditional Logic for DEV4
		A0150	Opening and Disabling Conditional Logic for DEV5
		A0060	Opening and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Opening and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Waiting for level LSL_382C in alarm and PU1382C feedback
T1600A	N2 Filtration		If (mode is <Auto> + 5 sec delay) or (mode is <Manual> and Command <N2 Filtration>) then go to step S1600
S1500	Blow2	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4

Step	Description	Action	Action Description
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Disabling Conditional Logic for DEV9
		A0095	Enabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Set Blow2 Time
		A0150	Start Blow2 Time
		A0160	Blow2 Time Message
T1700A	Blow3		If (Blow Time is over and mode is <Auto>) or (mode is <Manual> and Command <Blow3>) then go to step S1700
S1600	N2 Filtration	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Opening and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Opening and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Opening and Disabling Conditional Logic for DEV8
		A0090	Enabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13

Step	Description	Action	Action Description
		A0140	Set N2 Filtration Time
		A0150	Start N2 Filtration Time
		A0160	N2 Filtration Time Message
		A0170	Set Message
T1800B	Security Drain		If (N2 Filtration Time is over and mode is <Auto>) or (mode is <Manual> and Command <Security Drain>) and LSL_316A in alarm + 5 sec delay then go to step S1800
S1700	Blow3	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Opening and Disabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Enabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Set Blow3 Time
		A0150	Start Blow3 Time
		A0160	Blow3 Time Message
T1800A	Security Drain		If (Blow Time is over and mode is <Auto>) or (mode is <Manual> and Command <Security Drain>) then go to step S1800
S1800	Security Drain	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Opening and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5

Step	Description	Action	Action Description
		A0060	Opening and Disabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Set Security Drain Time
		A0150	Start Security Drain Time
		A0160	Security Drain Time Message
T1900A	Depressurisation		If (Security Drain Time is over and mode is <Auto>) or (mode is <Manual> and Command <Depressurisation>) then go to step S1900
S1900	Depressurisation	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Opening and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Opening and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Opening and Disabling Conditional Logic for DEV6
		A0070	Opening and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13

Step	Description	Action	Action Description
		A0140	Set Depressurization Time
		A0150	Start Depressurization Time
		A0160	Depressurization Time Message
		A0170	Set Parameter
T2000A	Spent Route Check		If (Depressurisation Time is over and mode is <Auto>) or (mode is <Manual> and Command <Spent Route Check>) then go to step S2000
S2000	Spent Route Check	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Opening and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Opening and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Opening and Disabling Conditional Logic for DEV6
		A0070	Opening and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Enabling Conditional Logic for DEV13
T2100A	Open V9		If (Operator selects <Drain to 1-1335B> and 1335B is in step <Filling Catalyst> and DEV13 is open) or (mode is <Manual> and Command <Open V9>) then go to step S2100
S2100	Open V9	A0010	Set Message: Reset Batches Filterd to Zero
		A0020	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0030	Closing and Disabling Conditional Logic for DEV2
		A0040	Opening DEV3
		A0050	Closing and Disabling Conditional Logic for DEV4

Step	Description	Action	Action Description
		A0060	Closing and Disabling Conditional Logic for DEV5
		A0070	Closing and Disabling Conditional Logic for DEV6
		A0080	Closing and Disabling Conditional Logic for DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Enabling Conditional Logic for DEV10
		A0120	Closing and Disabling Conditional Logic for DEV11
		A0130	Closing and Disabling Conditional Logic for DEV12
		A0140	Enabling Conditional Logic for DEV13
		A0150	Set Message: Set OP parameter to Open DEV10
T2200A	Discharge1		If (DEV10 is open + 5 sec delay and mode is <Auto>) or (DEV10 is open and mode is <Manual> and Command <Discharge1>) then go to step S2200
S2200	Discharge1	A0010	Set Task Pointer
		A0020	Stopping and Disabling Conditional Logic for DEV1
		A0030	Closing and Disabling Conditional Logic for DEV2
		A0040	Closing DEV3
		A0050	Enabling Conditional Logic for DEV4
		A0060	Closing and Disabling Conditional Logic for DEV5
		A0070	Closing and Disabling Conditional Logic for DEV6
		A0080	Closing and Disabling Conditional Logic for DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Opening and Disabling Conditional Logic for DEV10
		A0120	Opening and Disabling Conditional Logic for DEV11
		A0130	Closing and Disabling Conditional Logic for DEV12
		A0140	Enabling Conditional Logic for DEV13
		A0150	Set Discharge1 Time

Step	Description	Action	Action Description
		A0160	Start Discharge1 Time
		A0170	Discharge1Time Message
T2300A	Discharge2		If (Discharge Time is over and mode is <Auto>) or (mode is <Manual> and Command <Discharge2>) then go to step S2300
S2300	Discharge2	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Opening and Disabling Conditional Logic for DEV10
		A0110	Opening and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Set Discharge2 Time
		A0150	Start Discharge2 Time
		A0160	Discharge2 Time Message
T2400A	Discharge3		If (Discharge Time is over and mode is <Auto>) or (mode is <Manual> and Command <Discharge3>) then go to step S2400
S2400	Discharge3	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5

Step	Description	Action	Action Description
		A0060	Enabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Opening and Disabling Conditional Logic for DEV10
		A0110	Opening and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Set Discharge3 Time
		A0150	Start Discharge3 Time
		A0160	Discharge3 Time Message
T2400A	Standby Open		If (Discharge Time is over and mode is <Auto>) or (mode is <Manual> and Command <Standby Open>) then go to step S2400
S2500	Standby Open	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Opening and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Set Standby Open Time
		A0150	Start Standby Open Time

Step	Description	Action	Action Description
		A0160	Standby Open Time Message and Reset Drain 1-1335B OP
T2600A	Close V9		If (Standby Open Time is over and mode is <Auto> and Operator confirmation = <Close>) or (mode is <Manual> and Command <Close V9>) then go to step S2600
S2600	Close V9	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Enabling Conditional Logic for DEV13
T2700	Disable HM		If (DEV10 is closed + 5 sec delay and mode is <Auto>) or (mode is <Manual> and Command <Standby Off>) then go to step S2700
S2700	Disable HM	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8

Step	Description	Action	Action Description
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Disable HM 01 (Module Ownership Lost)
T2800	Release Devices		If no pending actions in step S2700 then go to step S2800
S2800	Release Devices	A0010	Release Private Device
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	Release Private Device
		A0100	Release Private Device
		A0110	Release Private Device
		A0120	Release Private Device
		A0130	Release Private Device
T9900	Complete		If no pending actions in S2800 then go to step S9900
S9900	Command Complete	A0010	Set Message: Filtration Command Complete
		A0020	Self Release the EM
END	Command Complete		True

Table 57: Filtration Sequence

8.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	RESET	Boolean	NA	False
OP002	Drain 1335B	Boolean	NA	False
OP003	MODE	Floating Point	NA	AUTO = 1 MAN = 0
OP004	Command	Named Set	NA	Standby ON
OP005	Open V_316A9	Boolean	NA	False
OP006	Close V_316A9	Boolean	NA	False
Tuning Parameters				
TP001	Discharge 3 Time	Floating Point	sec	5
TP002	Heating time	Floating Point	sec	0
TP003	Cleaning 1 time	Floating Point	sec	0
TP004	Precoat time	Floating Point	sec	0
TP005	Final Recirculation time	Floating Point	sec	0
TP006	Max. Batches	Floating Point	NA	0
TP007	Max. Filtration Pressure	Floating Point	barg	0.0
TP008	Safe Level Drain (LI_382C-1)	External Reference	%	HY1EQ_1382C/TP005_VALUE
TP009	Blow 1 time	Floating Point	sec	0
TP010	Security Drain time	Floating Point	sec	0
TP011	Drepressurization time	Floating Point	sec	0
TP012	Discharge 1 time	Floating Point	sec	0
TP013	Standby Open time	Floating Point	sec	0
TP014	Product Code	String	NA	
TP015	Filtered Batches	Floating Point	NA	0
TP016	LT_2382 Discharge SP	Floating Point	NA	0
TP017	LT_2382 Recycle SP	Floating Point	NA	0
TP018	N2 Filtration Time	Floating Point	sec	0
TP019	Cleaning 2 Time	Floating Point	sec	0
TP020	Cleaning 3 Time	Floating Point	sec	0
TP021	Blow 2 Time	Floating Point	sec	0

Parameter	Description	Type	Units	Default Value/Reference
TP022	Blow 3 Time	Floating Point	sec	0
TP023	Discharge 2 Time	Floating Point	sec	0

Table 58: Operating and Tuning Parameters

8.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		HY1EQ_1316A
OPR_CMD_MASK		3
DY_REF_1	Dynamic Reference	
DY_REF_2	Dynamic Reference	
DY_REF_3	Dynamic Reference	
DY_REF_4	Dynamic Reference	
DY_REF_5	Dynamic Reference	
DY_REF_6	Dynamic Reference	
DELAY_TMR	Floating Point	
STARTUP	String	HY1EQ_1300
REACTOR1	String	HY1EQ_1303
REACTOR2	String	HY1EQ_1382B
REACT_FILT	String	HY1EQ_1382C
NOT_SECU_DRN	Boolean	
PT_316A4	External Reference	HY1_PT1316A4/PV
LSL_382C	External Reference	HY1_LSL1282C/DI1/DISC_ACT
HOLD	External Reference	HY1EQ_1300/OP003_VALUE
DRAIN	Boolean	
PU_282C	External Reference	HY1_PU1282C/PV_D
STEP_TIME	Floating Point	0
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False
LSL_316AP	External Reference	HY1_LSL1316AP/DI1/DISC_ACT
LSH_316A	External Reference	HY1_LSH1316A/DI1/DISC_ACT
LSL_316A	External Reference	HY1_LSL1316A/DI1/DISC_ACT
LT_2382	External Reference	HY1_LT122382/PV
1335B	String	HY1EQ_1335B

Table 59: Configurable Parameters

8.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_HY_FILTER	Hydrogenation Area: Filtration EM

Table 60: Named Set Details

Details of states for NS_HY_FILTER named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Filtration

Table 61: Name Set State Details

9 EQUIPMENT MODULE CLASS DESIGN – EM_HY_FILTER_B (1-1316B)

9.1 Functional Description

EM_HY_FILTER_B is a class based equipment module that executes filtration operations. EM_HY_FILTER_B is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Process Filter Command and shall be released in Blow Command. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

9.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_HY_FILTER_B	HY1EQ_1316B
DEV1	_V12_IP_130_SE	HY1_V1116B1
DEV2	_V12_IP_130_SE	HY1_V1116B2
DEV3	_V12_IP_130_SE	HY1_V1116B3
DEV4	_V12_IP_130_SE	HY1_V1116B4
DEV5	_V12_IP_130_SE	HY1_V1116B5
DEV6	_V12_IP_130_SE	HY1_V1116B6
DEV7	_V12_IP_130_SE	HY1_V1016B8
DEV8	_V12_IP_130_SE	HY1_V1216B9
DEV9	_V12_IP_130_SE	HY1_V1216B10
DEV10	_V12_IP_130_SE	HY1_V1316BR

Table 62: Instances

9.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

9.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Common Hold Start	A0010	Store Fail Message
		A0020	Reset HOLD_REQ
		A0030	Save Previous Step Time

Step	Description	Action	Action Description
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
T9900A	Actions Complete or Time Out & Hold=True		Go to step H9900A
T9900B	Actions Complete or Time Out & Hold=False		Go to step H9900B
H9900A	Common Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Disable HM 01 (Module Ownership Lost)
		A0020	Release Devices Message
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices

Step	Description	Action	Action Description
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
		A0110	Release Private Devices
		A0120	Release Private Devices
		A0130	Reset Parameters
END1	Command Complete		True

Table 63: Hold Sequence

9.3.2

Command 01 – Process Filter

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Process Filter Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Acquire Private Device
		A0060	Acquire Private Device
		A0070	Acquire Private Device
		A0080	Acquire Private Device
		A0090	Acquire Private Device
		A0100	Acquire Private Device
		A0110	Acquire Private Device
		A0120	Reset Hold Request

Step	Description	Action	Action Description
T0100	Standby		Go to step S0100
T0200B	Fill		If Task Pointer = 2 then go to step S0200
T0300B	Heating		If Task Pointer = 3 then go to step S0300
T0400B	Cleaning		If Task Pointer = 4 then go to step S0400
T0500B	Refill		If Task Pointer = 5 then go to step S0500
T0600B	Precoat		If Task Pointer = 6 then go to step S0600
T0700B	Filtration		If Task Pointer = 7 then go to step S0700
T0800B	Drain Fwd		If Task Pointer = 8 then go to step S0800
T0900B	DrainTo82C		If Task Pointer = 9 then go to step S0900
T1000C	Drying		If Task Pointer = 10 then go to step S1000
T1100B	Security Drain		If Task Pointer = 11 then go to step S1100
T1200B	Depressurisation		If Task Pointer = 12 then go to step S1200
T1300B	Open V9		If Task Pointer = 13 then go to step S1300
T1400B	Discharge		If Task Pointer = 14 then go to step S1400
T1500B	Standby Open		If Task Pointer = 15 then go to step S1500
T1600B	Close		If Task Pointer = 16 then go to step S1600
T1700B	Close		If Task Pointer = 17 then go to step S1700
S0100	Standby	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Opening and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Opening and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10

Step	Description	Action	Action Description
T0200A	Fill		If (FILTER is in steps <Filtration/Final Filtration> and Polishing = ON) or (REACTOR4 is in step <Empty 1> and Heel Filtraion = ON) then go to step S0200.
S0200	Fill	A0010	Enabling Conditional Logic for DEV1
		A0020	Opening and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0120	Set Message: 5 sec delay
T0300A	Heating		If LSH_316B in alarm + 5 secs delay then go to step S0300.
S0300	Heating	A0010	Enabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8

Step	Description	Action	Action Description
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Set Heating Time
		A0120	Start Heating Time
		A0130	Heating Time Message
T0400A	Cleaning		It Heating Time is over then go to step S0400
S0400	Cleaning	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Opening and Disabling Conditional Logic for DEV8
		A0090	Opening DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Set Cleaning Time
		A0120	Start Cleaning Time
		A0130	Cleaning Time Message
		A0140	Closing DEV8
T0500A	Refill		If Cleaning Time is over and DEV8 is closed then go to step S0500
S0500	Refill	A0010	Enabling Conditional Logic for DEV1
		A0020	Opening and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5

Step	Description	Action	Action Description
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Set Message: 5 sec delay
T0600A	Precoat		If LSH_316B in alarm + 5 secs delay then go to step S0600.
S0600	Precoat	A0010	Enabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Set Precoat Time
		A0120	Start Precoat Time
		A0130	Blower Time Message
T0700A	Filtration		If Precoat Time is over then go to step S0700
S0700	Filtration	A0010	Enabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3

Step	Description	Action	Action Description
		A0040	Enabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Set Filtration Time
		A0120	Start Filtration Time
		A0130	Filtration Time Message
T0400C	Cleaning		If REACTOR4 is in step <Fill/Filtration/Recirculate Filter> and (Filtration Time Over or Operator Command = <Clean> or Max. Filtration Pressure is reached) then go to step S0400
T0800A	Drain Fwd		If REACTOR4 is in step <Drain Wait 1> then go to step S0800
T0900A	DrainTo82C		If REACTOR4 is in step <DrainFilter> and Heel Filtration = OFF or Operator Command = <Drain> then go to step S0900
S0800	Drain Fwd	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Opening and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Opening and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9

Step	Description	Action	Action Description
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Set Drain Fwd Time
		A0120	Start Drain Fwd Time
		A0130	Drain Fwd Time Message
T1000A	Drying		If Drain Forward Time is over then go to step S1000
S0900	DrainTo82C	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Opening and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Enabling Conditional Logic for DEV10
		A0110	Set Message: 5sec delay
		A0120	Set Message : LSL_316B alarm is active
T1000B	Drying		If LSL_316B in alarm + 5 secs delay then go to step S0600.
S1000	Drying	A0005	Unlatching Drain Bit 2
		A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6

Step	Description	Action	Action Description
		A0070	Opening and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Set Drying Time
		A0120	Start Drying Time
		A0130	Drying Time Message
T1100A	Security Drain		If Drying Time is over then go to step S1100
S1100	Security Drain	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Enabling Conditional Logic for DEV10
		A0110	Set Security Drain Time
		A0120	Start Security Drain Time
		A0130	Security Drain Time Message
T1200A	Depressurisation		If Security Drain Time is over then go to step S1200
S1200	Depressurisation	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Opening and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3

Step	Description	Action	Action Description
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Opening and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Opening and Disabling Conditional Logic for DEV10
		A0110	Set Depressurisation Time
		A0120	Start Depressurisation Time
		A0130	Depressurisation Time Message
T1300A	Open V9		If Depressurisation Time is over then go to step S1300
S1300	Open V9	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Opening and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Opening and Disabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Opening DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Set Message: Delay 5 sec
		A0120	Prompt: Open DEV14
T1400A	Discharge		If DEV8 is open + 5 sec delay then go to step S1400

Step	Description	Action	Action Description
S1400	Discharge	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Opening and Disabling Conditional Logic for DEV7
		A0080	Opening and Disabling Conditional Logic for DEV8
		A0090	Opening DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Set Discharge Time
		A0120	Start Discharge Time
		A0130	Discharge Time Message
T1500A	Standby Open		If Discharge Time is over then go to step S1500
S1500	Standby Open	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Opening and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Opening and Disabling Conditional Logic for DEV7
		A0080	Opening and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9

Step	Description	Action	Action Description
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Set Standby Open Time
		A0120	Start Standby Open Time
		A0130	Standby Open Time Message
T1600A	Close		If Standby Open Time is over and Operator Confirmation <Close> then go to step S1600
S1600	Close	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Opening and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Opening and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Set Message: Delay 5 sec
T1700A	Aux Off		If DEV8 is closed + 5 sec delay then go to step S1700
S1700	Aux Off	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Opening and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5

Step	Description	Action	Action Description
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Opening and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
T1800	Release Devices		If no pending actions in S1700 then go to step S1800
S1800	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	Release Private Device
		A0100	Release Private Device
		A0110	Release Private Device
		A0120	Release Private Device
T9900	Complete		If no pending actions in S1800 then go to step S9900
S9900	Complete	A0010	Set Message: Process Filter Command Complete
		A0020	Self Release the EM & Reset First Pass
END	Command Complete		True

Table 64: Process Filter Sequence

9.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Polishing	Boolean	NA	False
OP002	Heel Filtration	Boolean	NA	False
OP003	RESET	Boolean	NA	False
OP004	CLEANING	Boolean	NA	False
OP005	DRAIN	Boolean	NA	False
OP006	MODE	Boolean	NA	AUTO = 1 MAN = 0
OP007	OPEN	Boolean	NA	False
OP008	OPERATOR CONF OPEN/CLOSE	Named Set	NA	CLEAN
OP009	CLOSE	Boolean	NA	False
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Heating time	Floating Point	sec	0
TP003	Cleaning time	Floating Point	sec	0
TP004	Precoat time	Floating Point	sec	0
TP005	Security Drain time	Floating Point	sec	0
TP006	Drepress time	Floating Point	sec	0
TP007	Discharge time	Floating Point	sec	0
TP008	Standby Open time	Floating Point	sec	0
TP009	Max Filtration Pressure	Floating Point	barg	0
TP010	Filtration time	Floating Point	sec	0
TP011	Drain Forward Time	Floating Point	sec	0
TP012	Drying Time	Floating Point	sec	0
TP013	LATCH DRAIN BIT2	Floating Point	sec	0

Table 65: Operating and Tuning Parameters

9.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		HY1EQ_1316B
OPR_CMD_MASK		3
ESM	String	
FILTER	String	HY1EQ_1316A
LT_282C1	External Reference	HY1_LT1282C1/PV
DY_REF_1	Dynamic Reference	
DY_REF_2	Dynamic Reference	
DY_REF_3	Dynamic Reference	
LSH_316B	External Reference	HY1_LSL1216B/DI1/DISC_ACT
LSL_316B	External Reference	HY1_LSL1316B/DI1/DISC_ACT
LSL_282C	External Reference	HY1_LSL1282C/DI1/DISC_ACT
PT_1316C1	External Reference	HY1_PT1316B/AI1/PV
STEP_TIME	Floating Point	0
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False
HOLD	External Reference	HY1EQ_1300/OP003_VALUE
REACTOR4	String	HY1EQ_1382C
FLAG	External Reference	HY1EQ_1382C/FLAG
AFTER_DISCHARGE	Boolean	False

Table 66: Configurable Parameters

9.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_HY_FILTER_B	Hydrogenation Area: Filter EM

Table 67: Named Set Details

Details of states for NS_HY_FILTER_B named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Process Filter

Table 68: Name Set State Details

10 MODULE INTERLOCKS

10.1 Functional Description

Module	Name	Description	Expression	Force SP State	Delay On (s)
HY1_V1203D	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND <CR>('/HY1EQ_1382B/COMMAND_00001/S0200/ACTIVE.CV' = TRUE)	OPEN	NO
HY1_V1203D	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND <CR>('/HY1EQ_1382B/COMMAND_00001/S0200/ACTIVE.CV' != TRUE)	CLOSE	NO
HY1_V1382CBN	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND <CR>('/HY1_V1382CD/PV_D' = 'EDC_VLV_PV:CLOSED' AND <CR>('/HY1_V1116AY/PV_D' = 'EDC_VLV_PV:OPEN' AND <CR>('/HY1_V1016B8/PV_D' = 'EDC_VLV_PV:OPEN')) AND '^/PV_D' = 'EDC_VLV_PV:CLOSED'	OPEN	5
HY1_V1382CBN	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND <CR>'^/PV_D' = 'EDC_VLV_PV:OPEN'	CLOSE	30
HY1_V1382CBN	Condition 3	Auto Logic	('/TP01.CV' = TRUE) AND <CR>('/HY1_V1382CD/PV_D' = 'EDC_VLV_PV:CLOSED' OR <CR>('/HY1_V1116AY/PV_D' = 'EDC_VLV_PV:OPEN' OR <CR>('/HY1_V1016B8/PV_D' = 'EDC_VLV_PV:OPEN'))	CLOSE	NO
HY1_V1316ABN	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND <CR><CR>('/HY1_V1316A4/PV_D' = 'EDC_VLV_PV:CLOSED' AND <CR><CR>('/HY1_V1016B8/PV_D' = 'EDC_VLV_PV:OPEN') AND '^/PV_D' = 'EDC_VLV_PV:CLOSED'	OPEN	NO
HY1_V1316ABN	Condition 2	Auto Logic	'/TP01.CV' = TRUE AND <CR>'^/PV_D' = 'EDC_VLV_PV:OPEN'	CLOSE	5
HY1_V1316ABN	Condition 3	Auto Logic	('/TP01.CV' = TRUE) AND <CR>('/HY1_V1316A4/PV_D' = 'EDC_VLV_PV:CLOSED' OR <CR>('/HY1_V1016B8/PV_D' = 'EDC_VLV_PV:OPEN'))	CLOSE	30



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Module	Name	Description	Expression	Force SP State	Delay On (s)
HY1_V1316BR	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//HY1_LSH1282C1/DI1/DISC_ACT.CV' != TRUE AND <CR>'//HY1_LT122382/HI_HI_ALM.CV' != TRUE AND '//HY1EQ_1316B/TP013_VALUE' = TRUE)	OPEN	NO
HY1_V1316BR	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//HY1_LSH1282C1/DI1/DISC_ACT.CV' = TRUE OR <CR>'//HY1_LT122382/HI_HI_ALM.CV' = TRUE OR '//HY1EQ_1316B/TP013_VALUE' = FALSE)	CLOSE	NO
HY1_V1316BR	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND <CR>('//HY1_LSH1282C1/DI1/DISC_ACT.CV' != TRUE AND <CR>'//HY1_LT122382/HI_HI_ALM.CV' != TRUE)	OPEN	NO
HY1_V1316BR	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND <CR>('//HY1_LSH1282C1/DI1/DISC_ACT.CV' = TRUE OR <CR>'//HY1_LT122382/HI_HI_ALM.CV' = TRUE)	CLOSE	NO
HY1_V1216B9	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND '//HY1EQ_1316B/OP007_VALUE' = TRUE	OPEN	NO
HY1_V1216B9	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND '//HY1EQ_1316B/OP007_VALUE' = FALSE	CLOSE	NO
HY1_V1116B6	Condition 1	Auto Logic	('TP02.CV' = TRUE) AND<CR>('//HY1_V1316BR/PV_D' = '_EDC_VLV_PV:OPEN' AND <CR>'//HY1_V1116B5/PV_D' = '_EDC_VLV_PV:OPEN' AND '//HY1EQ_1316B/TP013_VALUE' = TRUE)	OPEN	NO
HY1_V1116B6	Condition 2	Auto Logic	('TP02.CV' = TRUE) AND<CR>('//HY1_V1316BR/PV_D' = '_EDC_VLV_PV:CLOSED' OR <CR>'//HY1_V1116B5/PV_D' = '_EDC_VLV_PV:CLOSED' OR '//HY1EQ_1316B/TP013_VALUE' = FALSE)	CLOSE	NO
HY1_V1116B6	Condition 3	Auto Logic	('TP03.CV' = TRUE) AND<CR>'//HY1_V1116B5/PV_D' = '_EDC_VLV_PV:OPEN'	OPEN	NO
HY1_V1116B6	Condition 4	Auto Logic	('TP03.CV' = TRUE) AND<CR>'//HY1_V1116B5/PV_D' = '_EDC_VLV_PV:CLOSED'	CLOSE	NO



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Module	Name	Description	Expression	Force SP State	Delay On (s)
HY1_V1116B6	Condition 5	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//HY1_V1116B4/PV_D' = ' _EDC_VLV_PV:OPEN';	OPEN	NO
HY1_V1116B6	Condition 6	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//HY1_V1116B4/PV_D' = ' _EDC_VLV_PV:CLOSED';	CLOSE	NO
HY1_V1116B5	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//HY1_LSH1282C1/DI1/DISC_ACT.CV' != TRUE AND <CR>'//HY1_LT122382/HI_HI_ALM.CV' != TRUE AND '//HY1EQ_1316B/TP013_VALUE' = TRUE)	OPEN	NO
HY1_V1116B5	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//HY1_LSH1282C1/DI1/DISC_ACT.CV' = TRUE OR<CR>'//HY1_LT122382/HI_HI_ALM.CV' = TRUE OR '//HY1EQ_1316B/TP013_VALUE' != TRUE)	CLOSE	NO
HY1_V1116B5	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND <CR>('//HY1_LSH1282C1/DI1/DISC_ACT.CV' != TRUE AND <CR>'//HY1_LT122382/HI_HI_ALM.CV' != TRUE)	OPEN	NO
HY1_V1116B5	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND <CR>('//HY1_LSH1282C1/DI1/DISC_ACT.CV' = TRUE OR<CR>'//HY1_LT122382/HI_HI_ALM.CV' = TRUE)	CLOSE	NO
HY1_V1116B4	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//HY1_PU1316A/PV_D.CV' = ' _EDC_MTR2_PV:RUNNING') OR <CR>('//HY1_PU1282C/PV_D.CV' = ' _EDC_MTR2_PV:RUNNING' AND '//HY1EQ_1382C/COMMAND_00001/S0700/ACTIVE.CV' = TRUE);	OPEN	NO
HY1_V1116B4	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//HY1_PU1316A/PV_D.CV' = ' _EDC_MTR2_PV:STOPPED') AND <CR>('//HY1_PU1282C/PV_D.CV' = ' _EDC_MTR2_PV:STOPPED' OR '//HY1EQ_1382C/COMMAND_00001/S0700/ACTIVE.CV' != TRUE))	CLOSE	NO
HY1_V1116B3	Condition 1	Auto Logic	'/TP01.CV' = TRUE AND '/PV_D.CV' = ' _EDC_VLV_PV:CLOSED'	OPEN	NO
HY1_V1116B3	Condition 2	Auto Logic	'/TP01.CV' = TRUE AND '/PV_D.CV' = ' _EDC_VLV_PV:OPEN'	CLOSE	NO



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HY1_V1116B3	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND <CR>('HY1_V1216B9/PV_D'=' _EDC_VLV_PV:OPEN' AND 'PV_D.CV' = ' _EDC_VLV_PV:CLOSED')	OPEN	NO
HY1_V1116B3	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND <CR>('PV_D.CV' = ' _EDC_VLV_PV:OPEN');	CLOSE	NO
HY1_V1116B3	Condition 5	Auto Logic	('TP02.CV' = TRUE) AND <CR>('HY1_V1216B9/PV_D'=' _EDC_VLV_PV:CLOSED');	CLOSE	NO
HY1_V1116B2	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'HY1_V1216B9/PV_D' = ' _EDC_VLV_PV:CLOSED';	OPEN	NO
HY1_V1116B2	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'HY1_V1216B9/PV_D' = ' _EDC_VLV_PV:OPEN';	CLOSE	NO
HY1_V1116B1	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_PU1316A/PV_D.CV' = ' _EDC_MTR2_PV:RUNNING') OR <CR>('HY1_PU1282C/PV_D.CV' ' _EDC_MTR2_PV:RUNNING' AND 'HY1EQ_1382C/COMMAND_00001/S0700/ACTIVE.CV' = TRUE);	OPEN	NO
HY1_V1116B1	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_PU1316A/PV_D.CV' = ' _EDC_MTR2_PV:STOPPED') AND <CR>('HY1_PU1282C/PV_D.CV' = ' _EDC_MTR2_PV:STOPPED' OR 'HY1EQ_1382C/COMMAND_00001/S0700/ACTIVE.CV' != TRUE))	CLOSE	NO
HY1_V1016B8	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'HY1EQ_1316B/TP013_VALUE' = TRUE	OPEN	NO
HY1_V1016B8	Condition 2	Auto Logic	('TP01.CV' = TRUE) OR <CR>'HY1EQ_1316B/TP013_VALUE' = FALSE	CLOSE	NO
HY1_V1382CR	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND<CR>('HY1_LSH1282C1/DI1/DISC_ACT.CV' != TRUE AND<CR>'HY1_LT122382/AI1/HI_HI_ACT.CV' != TRUE AND<CR>'HY1EQ_1316A/DRAIN.CV' = TRUE);	OPEN	NO
HY1_V1382CR	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND<CR>('HY1_LSH1282C1/DI1/DISC_ACT.CV' = TRUE OR	CLOSE	NO



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Module	Name	Description	Expression	Force SP State	Delay On (s)
			<CR>'//HY1_LT122382/AI1/HI_HI_ACT.CV' = TRUE OR '//HY1EQ_1316A/DRAIN.CV' = FALSE);		
HY1_V1382CR	Condition 3	Auto Logic	('//TP02.CV' = TRUE) AND<CR>'//HY1_LSH1282C1/DI1/DISC_ACT.CV' != TRUE AND<CR>'//HY1_LT122382/AI1/HI_HI_ACT.CV' != TRUE)	OPEN	NO
HY1_V1382CR	Condition 4	Auto Logic	('//TP02.CV' = TRUE) AND<CR>'//HY1_LSH1282C1/DI1/DISC_ACT.CV' = TRUE OR <CR>'//HY1_LT122382/AI1/HI_HI_ACT.CV' = TRUE)	CLOSE	NO
HY1_V1357CK	Condition 1	Auto Logic	('//TP01.CV' = TRUE) AND <CR>'//HY1EQ_1316A/OP002_VALUE.CV' = TRUE)	OPEN	NO
HY1_V1357CK	Condition 2	Auto Logic	('//TP01.CV' = TRUE) AND <CR>'//HY1EQ_1316A/OP002_VALUE.CV' != TRUE)	CLOSE	NO
HY1_V1316AV3C	Condition 1	Auto Logic	('//PV_D.CV' = '_EDC_VLV_PV:CLOSED') AND <CR>'//TP01.CV' = TRUE)	OPEN	5
HY1_V1316AV3C	Condition 2	Auto Logic	('//PV_D.CV' = '_EDC_VLV_PV:OPEN') AND <CR>'//TP01.CV' = TRUE)	CLOSE	5
HY1_V1316AV3C	Condition 3	Auto Logic	('//TP02.CV' = TRUE) AND <CR>'//HY1_V1316A9/PV_D' = '_EDC_VLV_PV:OPEN' AND <CR>'//PV_D.CV' = '_EDC_VLV_PV:CLOSED'	OPEN	10
HY1_V1316AV3C	Condition 4	Auto Logic	('//TP02.CV' = TRUE) AND <CR>'//PV_D.CV' = '_EDC_VLV_PV:OPEN'	CLOSE	10
HY1_V1316AV3C	Condition 5	Auto Logic	('//TP02.CV' = TRUE) AND <CR>'//HY1_V1316A9/PV_D' = '_EDC_VLV_PV:CLOSED'	CLOSE	NO
HY1_V1316AV3B	Condition 1	Auto Logic	('//PV_D.CV' = '_EDC_VLV_PV:CLOSED') AND <CR>'//TP01.CV' = TRUE)	OPEN	5
HY1_V1316AV3B	Condition 2	Auto Logic	('//PV_D.CV' = '_EDC_VLV_PV:OPEN') AND <CR>'//TP01.CV' = TRUE)	CLOSE	5



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Module	Name	Description	Expression	Force SP State	Delay On (s)
HY1_V1316AV3B	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND <CR>'//HY1_V1316A9/PV_D' = '_EDC_VLV_PV:OPEN' AND<CR>'//PV_D.CV' = '_EDC_VLV_PV:CLOSED'	OPEN	10
HY1_V1316AV3B	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND <CR>'//PV_D.CV' = '_EDC_VLV_PV:OPEN'	CLOSE	10
HY1_V1316AV3B	Condition 5	Auto Logic	('TP02.CV' = TRUE) AND <CR>'//HY1_V1316A9/PV_D' = '_EDC_VLV_PV:CLOSED'	CLOSE	NO
HY1_V1316AV3A	Condition 1	Auto Logic	('PV_D.CV' = '_EDC_VLV_PV:CLOSED') AND <CR>('TP01.CV' = TRUE)	OPEN	5
HY1_V1316AV3A	Condition 2	Auto Logic	('PV_D.CV' = '_EDC_VLV_PV:OPEN') AND <CR>('TP01.CV' = TRUE)	CLOSE	5
HY1_V1316AV3A	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND <CR>'//HY1_V1316A9/PV_D' = '_EDC_VLV_PV:OPEN' AND<CR>'//PV_D.CV' = '_EDC_VLV_PV:CLOSED'	OPEN	10
HY1_V1316AV3A	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND <CR>'//PV_D.CV' = '_EDC_VLV_PV:OPEN'	CLOSE	10
HY1_V1316AV3A	Condition 5	Auto Logic	('TP02.CV' = TRUE) AND <CR>'//HY1_V1316A9/PV_D' = '_EDC_VLV_PV:CLOSED'	CLOSE	NO
HY1_V1316A9	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//HY1EQ_1316A/OP005_VALUE.CV' = TRUE)	OPEN	NO
HY1_V1316A9	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//HY1EQ_1316A/OP005_VALUE.CV' != TRUE)	CLOSE	NO
HY1_V1316A6	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND<CR>('//HY1EQ_1316A/DRAIN.CV' = TRUE)	OPEN	NO
HY1_V1316A6	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND<CR>('//HY1EQ_1316A/DRAIN.CV' = FALSE);	CLOSE	NO
HY1_V1316A6	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND <CR>'//HY1_V1316A4/PV_D' = '_EDC_VLV_PV:OPEN' AND <CR>('//HY1_V1316AV3A/PV_D' = '_EDC_VLV_PV:OPEN' OR<CR>'//HY1_V1316AV3B/PV_D' =	OPEN	NO



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Module	Name	Description	Expression	Force SP State	Delay On (s)
			'_EDC_VLV_PV:OPEN' OR <CR>'//HY1_V1316AV3C/PV_D' = '_EDC_VLV_PV:OPEN'));		
HY1_V1316A6	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND <CR>'//HY1_V1316A4/PV_D' = '_EDC_VLV_PV:CLOSED' OR <CR>'//HY1_V1316AV3A/PV_D' = '_EDC_VLV_PV:CLOSED' AND <CR>'//HY1_V1316AV3B/PV_D' = '_EDC_VLV_PV:CLOSED' AND <CR>'//HY1_V1316AV3C/PV_D' = '_EDC_VLV_PV:CLOSED'));	CLOSE	NO
HY1_V1316A6	Condition 5	Auto Logic	('TP03.CV' = TRUE) AND <CR>'//HY1_V1316AV3A/PV_D' = '_EDC_VLV_PV:OPEN' AND <CR>'//HY1_V1316A3/PV_D' = '_EDC_VLV_PV:OPEN'));	OPEN	NO
HY1_V1316A6	Condition 6	Auto Logic	('TP03.CV' = TRUE) AND <CR>'//HY1_V1316AV3A/PV_D' = '_EDC_VLV_PV:CLOSED' OR <CR>'//HY1_V1316A3/PV_D' = '_EDC_VLV_PV:CLOSED'));	CLOSE	NO
HY1_V1316A6	Condition 7	Auto Logic	('TP04.CV' = TRUE AND '//HY1_V1316AV3B/PV_D' = '_EDC_VLV_PV:OPEN' AND '//HY1_V1316A3/PV_D' = '_EDC_VLV_PV:OPEN') OR (('TP05.CV' = TRUE AND '//HY1_V1316AV3C/PV_D' = '_EDC_VLV_PV:OPEN' AND '//HY1_V1316A3/PV_D' = '_EDC_VLV_PV:OPEN'));	OPEN	NO
HY1_V1316A6	Condition 8	Auto Logic	((('TP04.CV' = TRUE) AND (('HY1_V1316AV3B/PV_D' != '_EDC_VLV_PV:OPEN' OR '//HY1_V1316A3/PV_D' != '_EDC_VLV_PV:OPEN')) OR ((('TP05.CV' = TRUE) AND (('HY1_V1316AV3C/PV_D' != '_EDC_VLV_PV:OPEN' OR '//HY1_V1316A3/PV_D' != '_EDC_VLV_PV:OPEN')));	CLOSE	NO
HY1_V1316A4	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND<CR>'//HY1_LT122382/PV' > '//HY1EQ_1316A/TP016_VALUE.CV'	OPEN	NO



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Module	Name	Description	Expression	Force SP State	Delay On (s)
HY1_V1316A4	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND<CR>('HY1_LT122382/PV' < 'HY1EQ_1316A/TP017_VALUE.CV')	CLOSE	NO
HY1_V1316A3	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND<CR>('HY1_LT122382/PV' <= 'HY1EQ_1316A/TP017_VALUE.CV')	OPEN	NO
HY1_V1316A3	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND<CR>('HY1_LT122382/PV' > 'HY1EQ_1316A/TP016_VALUE.CV')	CLOSE	NO
HY1_V1316A3	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND<CR>('HY1_LT122382/A11/HI_HI_ACT.CV' != TRUE AND <CR>('HY1_LSH1282C1/DI1/DISC_ACT.CV' != TRUE AND<CR>('HY1EQ_1316A/DRAIN.CV' = TRUE));	OPEN	NO
HY1_V1316A3	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND<CR>('HY1_LT122382/A11/HI_HI_ACT.CV' = TRUE OR <CR>('HY1_LSH1282C1/DI1/DISC_ACT.CV' = TRUE OR 'HY1EQ_1316A/DRAIN.CV' != TRUE));	CLOSE	NO
HY1_V1316A3	Condition 5	Auto Logic	('TP03.CV' = TRUE) AND<CR>('HY1_LSH1282C1/DI1/DISC_ACT.CV' != TRUE AND<CR>('HY1_LT122382/A11/HI_HI_ACT.CV' != TRUE));	OPEN	NO
HY1_V1316A3	Condition 6	Auto Logic	('TP03.CV' = TRUE) AND<CR>('HY1_LSH1282C1/DI1/DISC_ACT.CV' = TRUE OR<CR>('HY1_LT122382/A11/HI_HI_ACT.CV' = TRUE));	CLOSE	NO
HY1_V1316A1	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND<CR>('HY1_PU1282C/PV_D' = 'EDC_MTR2_PV:RUNNING')	OPEN	NO
HY1_V1316A1	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND<CR>('HY1_PU1282C/PV_D' != 'EDC_MTR2_PV:RUNNING')	CLOSE	NO
HY1_V1382CF3	Condition 1	Auto Logic	((('TP01.CV' = TRUE) AND <CR>('HY1_PU1282B/PV_D' = 'EDC_MTR2_PV:RUNNING' AND<CR>('HY1EQ_1302_82A/COMMAND_00001/S0400/ACTIVE.C V' != TRUE));	OPEN	NO
HY1_V1382CF3	Condition 2	Auto Logic	((('TP01.CV' = TRUE) AND <CR>('HY1_PU1282B/PV_D' = 'EDC_MTR2_PV:STOPPED'	CLOSE	NO



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			OR<CR>'//HY1EQ_1302_82A/COMMAND_00001/S0400/ACTIVE.CV' = TRUE));		
HY1_V1381AF	Condition 1	Auto Logic	((('/TP01.CV' = TRUE) AND <CR>'//HY1_PU1282B/PV_D' = '_EDC_MTR2_PV:RUNNING' AND<CR>'//HY1EQ_1302_82A/COMMAND_00001/S0400/ACTIVE.C V' = TRUE));	OPEN	NO
HY1_V1381AF	Condition 2	Auto Logic	((('/TP01.CV' = TRUE) AND <CR>'//HY1_PU1282B/PV_D' = '_EDC_MTR2_PV:STOPPED' OR<CR>'//HY1EQ_1302_82A/COMMAND_00001/S0400/ACTIVE.CV' != TRUE));	CLOSE	NO
HY1_V1381ABN	Condition 1	Auto Logic	('//TP01.CV' = TRUE) AND <CR>'//HY1_V1282BF1/PV_D' = '_EDC_VLV_PV:OPEN' AND<CR>'//HY1_V1381AF/PV_D' = '_EDC_VLV_PV:CLOSED' AND <CR>'//HY1_V1382CF3/PV_D' = '_EDC_VLV_PV:OPEN')AND<CR>'//PV_D.CV' = '_EDC_VLV_PV:CLOSED'	OPEN	5
HY1_V1381ABN	Condition 2	Auto Logic	('//TP01.CV' = TRUE) AND <CR>'//PV_D.CV' = '_EDC_VLV_PV:OPEN'	CLOSE	30
HY1_V1381ABN	Condition 3	Auto Logic	('//TP01.CV' = TRUE) AND <CR>'//HY1_V1282BF1/PV_D' = '_EDC_VLV_PV:CLOSED' OR<CR>'//HY1_V1381AF/PV_D' = '_EDC_VLV_PV:OPEN' OR<CR>'//HY1_V1382CF3/PV_D' = '_EDC_VLV_PV:CLOSED')	CLOSE	NO
HY1_V1282BF1	Condition 1	Auto Logic	((('/TP01.CV' = TRUE) AND <CR>'//HY1_PU1282B/PV_D' = '_EDC_MTR2_PV:RUNNING' AND<CR>'//HY1EQ_1302_82A/COMMAND_00001/S0400/ACTIVE.C V' = TRUE));	OPEN	NO
HY1_V1282BF1	Condition 2	Auto Logic	((('/TP01.CV' = TRUE) AND <CR>'//HY1_PU1282B/PV_D' = '_EDC_MTR2_PV:STOPPED' OR<CR>'//HY1EQ_1302_82A/COMMAND_00001/S0400/ACTIVE.CV' != TRUE));	CLOSE	NO



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HY1_V1182BBN	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//HY1_V1282BF1/PV_D' = '_EDC_VLV_PV:CLOSED' AND<CR>'//HY1_V1381AF/PV_D' = '_EDC_VLV_PV:CLOSED' AND <CR>'//HY1_V1382CF3/PV_D' = '_EDC_VLV_PV:OPEN') AND<CR>'//PV_D.CV' = '_EDC_VLV_PV:CLOSED'	OPEN	5
HY1_V1182BBN	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//PV_D.CV' = '_EDC_VLV_PV:OPEN'	CLOSE	30
HY1_V1182BBN	Condition 3	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//HY1_V1282BF1/PV_D' != '_EDC_VLV_PV:CLOSED' OR<CR>'//HY1_V1381AF/PV_D' != '_EDC_VLV_PV:CLOSED' OR <CR>'//HY1_V1382CF3/PV_D' != '_EDC_VLV_PV:OPEN')	CLOSE	NO
HY1_V1203S	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//HY1_V1203/PV_D' = '_EDC_VLV_PV:OPEN' AND <CR>'//HY1_V1203WR/PV_D' = '_EDC_VLV_PV:CLOSED' AND <CR>'//HY1_V1203E/PV_D' = '_EDC_VLV_PV:CLOSED')	OPEN	NO
HY1_V1203S	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//HY1_V1203/PV_D' != '_EDC_VLV_PV:OPEN' OR <CR>'//HY1_V1203WR/PV_D' != '_EDC_VLV_PV:CLOSED' OR <CR>'//HY1_V1203E/PV_D' != '_EDC_VLV_PV:CLOSED')	CLOSE	NO
HY1_V1335BS	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//HY1_LSL1135B/DI1/DISC_ACT' != TRUE AND 'HY1_TT1135B/PV.CV' < ('HY1EQ_1335B/OP004_VALUE.CV'))	OPEN	NO
HY1_V1335BS	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//HY1_LSL1135B/DI1/DISC_ACT' = TRUE OR 'HY1_TT1135B/PV.CV' >= ('HY1EQ_1335B/OP004_VALUE.CV' + 2))	CLOSE	NO
HY1_V1235BD	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//HY1_V1235/PV_D' = '_EDC_VLV_PV:OPEN' AND 'HY1EQ_1303/COMMAND_00001/S0400/ACTIVE.CV' = TRUE)	OPEN	NO



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Module	Name	Description	Expression	Force SP State	Delay On (s)
HY1_V1235BD	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND ('/HY1_V1235/PV_D' != ' _EDC_VLV_PV:OPEN' OR '/HY1EQ_1303/COMMAND_00001/S0400/ACTIVE.CV' != TRUE)	CLOSE	NO
HY1_V1235BBN	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND <CR>('/HY1_V1235BD/PV_D' = ' _EDC_VLV_PV:CLOSED') AND '^/PV_D.CV' = ' _EDC_VLV_PV:CLOSED'	OPEN	5
HY1_V1235BBN	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND <CR>'^/PV_D.CV' = ' _EDC_VLV_PV:OPEN'	CLOSE	30
HY1_V1235BBN	Condition 3	Auto Logic	('/TP01.CV' = TRUE) AND <CR>('/HY1_V1235BD/PV_D' = ' _EDC_VLV_PV:OPEN')	CLOSE	NO
HY1_V1035BF	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND ('/HY1_LSH1135B/DI1/DISC_ACT' != TRUE)	OPEN	NO
HY1_V1035BF	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND ('/HY1_LSH1135B/DI1/DISC_ACT' = TRUE)	CLOSE	NO
HY1_V1235BN	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND <CR>('/HY1_V1135/PV_D' = ' _EDC_VLV_PV:CLOSED') AND '^/PV_D.CV' = ' _EDC_VLV_PV:CLOSED'	OPEN	5
HY1_V1235BN	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND <CR>'^/PV_D.CV' = ' _EDC_VLV_PV:OPEN'	CLOSE	30
HY1_V1235BN	Condition 3	Auto Logic	('/TP01.CV' = TRUE) AND <CR>('/HY1_V1135/PV_D' = ' _EDC_VLV_PV:OPEN')	CLOSE	NO
HY1_V1235AS	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND <CR>('/HY1_LSL1135A/DI1/DISC_ACT' != TRUE AND '/HY1_TT1135A/PV.CV' < ('/HY1EQ_1335A/OP004_VALUE.CV'))	OPEN	NO
HY1_V1235AS	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND <CR>('/HY1_LSL1135A/DI1/DISC_ACT' = TRUE OR '/HY1_TT1135A/PV.CV' >= ('/HY1EQ_1335A/OP004_VALUE.CV' + 2))	CLOSE	NO
HY1_V1235A	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND ('/HY1_LSH1135A/DI1/DISC_ACT' != TRUE)	OPEN	NO



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Module	Name	Description	Expression	Force SP State	Delay On (s)
HY1_V1235A	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND ('/HY1_LSH1135A/DI1/DISC_ACT' = TRUE)	CLOSE	NO
HY1_V1135	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('/HY1_V1235/PV_D' = 'EDC_VLV_PV:OPEN' AND '/HY1EQ_1303/COMMAND_00001/S0400/ACTIVE.CV' = TRUE)	OPEN	NO
HY1_V1135	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('/HY1_V1235/PV_D' != 'EDC_VLV_PV:OPEN' OR '/HY1EQ_1303/COMMAND_00001/S0400/ACTIVE.CV' != TRUE)	CLOSE	NO
HY1_V1235	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND<CR>('/HY1EQ_1335A/OWNER_ID.CV' = "HY1EQ_1303_82A" AND ('/HY1EQ_1335A/COMMAND_00002/S0200/ACTIVE.CV' = TRUE OR '/HY1EQ_1335A/COMMAND_00002/S0300/ACTIVE.CV' = TRUE)) OR <CR> ('/HY1EQ_1335B/OWNER_ID.CV' = "HY1EQ_1303_82A" AND ('/HY1EQ_1335B/COMMAND_00002/S0200/ACTIVE.CV' = TRUE OR '/HY1EQ_1335B/COMMAND_00002/S0300/ACTIVE.CV' = TRUE)))	OPEN	NO
HY1_V1235	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('/HY1EQ_1335A/OWNER_ID.CV' = "HY1EQ_1303_82A" AND ('/HY1EQ_1335A/COMMAND_00002/S0200/ACTIVE.CV' != TRUE AND '/HY1EQ_1335A/COMMAND_00002/S0300/ACTIVE.CV' != TRUE)) OR <CR> ('/HY1EQ_1335B/OWNER_ID.CV' = "HY1EQ_1303_82A" AND ('/HY1EQ_1335B/COMMAND_00002/S0200/ACTIVE.CV' != TRUE AND '/HY1EQ_1335B/COMMAND_00002/S0300/ACTIVE.CV' != TRUE)))	CLOSE	NO
HY1_V1203VA	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('/HY1_PT1203/PV' >= 0.2)	OPEN	NO
HY1_V1203VA	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('/HY1_PT1203/PV' < 0.2)	CLOSE	NO



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Module	Name	Description	Expression	Force SP State	Delay On (s)
HY1_V1203V1	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('/HY1EQ_1300/OWNER_ID.CV' != "(None)")	OPEN	NO
HY1_V1203V1	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('/HY1EQ_1300/OWNER_ID.CV' = "(None)")	CLOSE	NO
HY1_V1203V1	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND (('HY1_PT1203/PV.CV') >= ('HY1EQ_1303/TP006_VALUE.CV'))	OPEN	NO
HY1_V1203V1	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND (('HY1_PT1203/PV.CV') < ('HY1EQ_1303/TP006_VALUE.CV' - 5))	CLOSE	NO
HY1_V1203H	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('/HY1_PT1203/PID1/HI_HI_ACT.CV' != TRUE) AND <CR>('/HY1_PSH1203/DI1/DISC_ACT' != TRUE) AND <CR>('/HY1_RD12031/DI1/DISC_ACT.CV' != TRUE) AND <CR>('/HY1_V1203F/PV_D' = 'EDC_VLV_PV:CLOSED') AND <CR>('/HY1_V1203D/PV_D' = 'EDC_VLV_PV:CLOSED') AND <CR>('/HY1_V1203VA/PV_D' = 'EDC_VLV_PV:CLOSED') AND <CR>('/HY1_V1203V1/PV_D' = 'EDC_VLV_PV:CLOSED')	OPEN	NO
HY1_V1203H	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('/HY1_PT1203/PID1/HI_HI_ACT.CV' = TRUE) OR <CR>('/HY1_PSH1203/DI1/DISC_ACT' = TRUE) OR <CR>('/HY1_RD12031/DI1/DISC_ACT.CV' = TRUE) OR <CR>('/HY1_V1203F/PV_D' != 'EDC_VLV_PV:CLOSED') OR <CR>('/HY1_V1203D/PV_D' != 'EDC_VLV_PV:CLOSED') OR <CR>('/HY1_V1203VA/PV_D' != 'EDC_VLV_PV:CLOSED') OR <CR>('/HY1_V1203V1/PV_D' != 'EDC_VLV_PV:CLOSED')	CLOSE	NO
HY1_V1203F	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('/HY1_LSH1203/DI1/DISC_ACT.CV' != TRUE AND '/HY1_V1203D/PV_D' = 'EDC_VLV_PV:CLOSED')	OPEN	NO
HY1_V1203F	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('/HY1_LSH1203/DI1/DISC_ACT.CV' = TRUE OR '/HY1_V1203D/PV_D' != 'EDC_VLV_PV:CLOSED')	CLOSE	NO



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Module	Name	Description	Expression	Force SP State	Delay On (s)
HY1_V1382AMPS	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_TT1002A/PV.CV' < 'HY1EQ_1302_82A/TP004_VALUE.CV' AND <CR>'HY1_V1002LPS/PV_D' = 'EDC_VLV_PV:OPEN');	OPEN	5
HY1_V1382AMPS	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_TT1002A/PV.CV' >= ('HY1EQ_1302_82A/TP004_VALUE.CV' + 2))	CLOSE	NO
HY1_V1002LPS	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND ('HY1_LSL1002A/DI1/DISC_ACT' != TRUE)	OPEN	NO
HY1_V1002LPS	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND ('HY1_LSL1002A/DI1/DISC_ACT' = TRUE)	CLOSE	NO
HY1_V1001BN	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_V1002F/PV_D' = 'EDC_VLV_PV:OPEN' AND <CR>'HY1_V1001F/PV_D' = 'EDC_VLV_PV:CLOSED') AND <CR>('PV_D' = 'EDC_VLV_PV:CLOSED')	OPEN	5
HY1_V1001BN	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('PV_D' = 'EDC_VLV_PV:OPEN')	CLOSE	30
HY1_V1001BN	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_V1002F/PV_D' = 'EDC_VLV_PV:CLOSED' OR <CR>'HY1_V1001F/PV_D' = 'EDC_VLV_PV:OPEN')	CLOSE	NO
HY1_V131078TW	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_LSH1078TW/DI1/DISC_ACT.CV' != TRUE)	OPEN	NO
HY1_V131078TW	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_LSH1078TW/DI1/DISC_ACT.CV' = TRUE)	CLOSE	NO
HY1_PU1350B	Condition 1	Auto Logic	'MODE.CV' = CAS AND<CR>('HY1_PU1316A/PV_D.CV' = 'EDC_MTR2_PV:RUNNING' OR 'HY1_PU1282C/PV_D.CV' = 'EDC_MTR2_PV:RUNNING') AND<CR>('HY1_V1116B4/PV_D.CV' = 'EDC_VLV_PV:OPEN' OR 'HY1_V1016B8/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND<CR>'HY1_FT1381B/AI1/LO_ACT.CV' != TRUE	START	5
HY1_PU1350B	Condition 2	Auto Logic	'MODE.CV' = CAS AND<CR>'HY1EQ_1300/OP003_VALUE.CV' = 1	STOP	NO



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Module	Name	Description	Expression	Force SP State	Delay On (s)
HY1_PU1001	Condition 1	Auto Logic	(('/TP01.CV' = TRUE) AND <CR>('/HY1_V1001F/PV_D' = '_EDC_VLV_PV:OPEN' AND<CR>('/OWNER_ID.CV' = "HY1EQ_1335A" AND '/HY1_V1235A/PV_D' = '_EDC_VLV_PV:OPEN') OR ('/OWNER_ID.CV' = "HY1EQ_1335B" AND '/HY1_V1035BF/PV_D' = '_EDC_VLV_PV:OPEN')) AND<CR>'/HY1EQ_1300/OP006_VALUE.CV' = TRUE);	START	5
HY1_PU1001	Condition 2	Auto Logic	(('/TP01.CV' = TRUE) AND <CR>('/HY1_V1001F/PV_D' != '_EDC_VLV_PV:OPEN' OR <CR>('/OWNER_ID.CV' = "HY1EQ_1335A" AND '/HY1_V1235A/PV_D' != '_EDC_VLV_PV:OPEN') OR ('/OWNER_ID.CV' = "HY1EQ_1335B" AND '/HY1_V1035BF/PV_D' != '_EDC_VLV_PV:OPEN')) OR<CR>'/HY1EQ_1300/OP006_VALUE.CV' = FALSE);	STOP	NO
HY1_PU1282C	Condition 1	Auto Logic	(('/TP01.CV' = TRUE) AND <CR>('/HY1_V1382CD/PV_D' = '_EDC_VLV_PV:OPEN') AND <CR>('/HY1_LSL1282C/DI1/DISC_ACT.CV' != TRUE) AND<CR>('/HY1EQ_1316A/COMMAND_00001/S0300/ACTIVE.CV' = TRUE OR<CR>'/HY1EQ_1316A/COMMAND_00001/S0400/ACTIVE.CV' = TRUE OR<CR>'/HY1EQ_1316A/COMMAND_00001/S0800/ACTIVE.CV' = TRUE OR<CR>'/HY1EQ_1316A/COMMAND_00001/S0900/ACTIVE.CV' = TRUE OR<CR>'/HY1EQ_1316A/COMMAND_00001/S1000/ACTIVE.CV' = TRUE OR<CR>'/HY1EQ_1316A/COMMAND_00001/S1200/ACTIVE.CV' = TRUE);	START	5
HY1_PU1282C	Condition 2	Auto Logic	(('/TP01.CV' = TRUE) AND <CR>('/HY1_V1382CD/PV_D' = '_EDC_VLV_PV:CLOSED') OR	STOP	NO



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Module	Name	Description	Expression	Force SP State	Delay On (s)
			<code><CR>('/HY1_LSL1282C/DI1/DISC_ACT.CV' = TRUE) OR<CR>('/HY1EQ_1316A/COMMAND_00001/S0300/ACTIVE.CV' = TRUE AND<CR>('/HY1EQ_1316A/COMMAND_00001/S0400/ACTIVE.CV' = TRUE AND<CR>('/HY1EQ_1316A/COMMAND_00001/S0800/ACTIVE.CV' = TRUE AND<CR>('/HY1EQ_1316A/COMMAND_00001/S0900/ACTIVE.CV' = TRUE AND<CR>('/HY1EQ_1316A/COMMAND_00001/S1000/ACTIVE.CV' = TRUE AND<CR>('/HY1EQ_1316A/COMMAND_00001/S1200/ACTIVE.CV' = TRUE))</code>		
HY1_PU1282C	Condition 3	Auto Logic	<code>('TP02.CV' = TRUE) AND <CR>('/HY1_V1382CD/PV_D' = '_EDC_VLV_PV:OPEN' AND <CR>('/HY1_LSL1282C/DI1/DISC_ACT.CV' != TRUE AND<CR>('/HY1EQ_1316A/COMMAND_00001/S1200/ACTIVE.CV' = TRUE OR<CR>('/HY1EQ_1316A/COMMAND_00001/S1400/ACTIVE.CV' = TRUE)</code>	START	NO
HY1_PU1282C	Condition 4	Auto Logic	<code>('TP02.CV' = TRUE) AND <CR>('/HY1_V1382CD/PV_D' = '_EDC_VLV_PV:CLOSED' OR <CR>('/HY1_LSL1282C/DI1/DISC_ACT.CV' = TRUE OR<CR>('/HY1EQ_1316A/COMMAND_00001/S1200/ACTIVE.CV' = TRUE AND<CR>('/HY1EQ_1316A/COMMAND_00001/S1400/ACTIVE.CV' = TRUE)</code>	STOP	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
HY1_PU1282C	Condition 5	Auto Logic	('TP04.CV' = TRUE) AND <CR>'//HY1_V1382CD/PV_D' = '_EDC_VLV_PV:OPEN' AND <CR>'//HY1_V1016B8/PV_D' = '_EDC_VLV_PV:OPEN';	START	5
HY1_PU1282C	Condition 6	Auto Logic	('TP04.CV' = TRUE) AND <CR>'//HY1_V1382CD/PV_D' = '_EDC_VLV_PV:CLOSED' OR <CR>'//HY1_V1016B8/PV_D' = '_EDC_VLV_PV:CLOSED';	STOP	NO
HY1_PU1282C	Condition 7	Auto Logic	('TP03.CV' = TRUE) AND '//HY1_V1382CD/PV_D' = '_EDC_VLV_PV:OPEN';	START	NO
HY1_PU1282C	Condition 8	Auto Logic	('TP03.CV' = TRUE) AND '//HY1_V1382CD/PV_D' = '_EDC_VLV_PV:CLOSED';	STOP	NO
HY1_AG1282C	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//HY1_LT122382/PV.CV' >= 'HY1EQ_1382C/TP008_VALUE.CV')	START	NO
HY1_AG1282C	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//HY1_LT122382/PV.CV' < 'HY1EQ_1382C/TP008_VALUE.CV')	STOP	NO
HY1_PU1316A	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//HY1_LSL1316AP/DI1/DISC_ACT.CV' != TRUE AND <CR>'//HY1_V1316A3/PV_D' = '_EDC_VLV_PV:CLOSED' AND <CR>'//HY1_V1316A4/PV_D' = '_EDC_VLV_PV:OPEN'	START	NO
HY1_PU1316A	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//HY1_LSL1316AP/DI1/DISC_ACT.CV' = TRUE OR<CR>'//HY1_V1316A3/PV_D' = '_EDC_VLV_PV:OPEN' OR <CR>'//HY1_V1316A4/PV_D' = '_EDC_VLV_PV:CLOSED')	STOP	NO
HY1_PU1316A	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND <CR>'//HY1_LSL1316AP/DI1/DISC_ACT.CV' != TRUE AND <CR>'//HY1_V1316A4/PV_D' = '_EDC_VLV_PV:OPEN';	START	NO
HY1_PU1316A	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND <CR>'//HY1_LSL1316AP/DI1/DISC_ACT.CV' = TRUE OR <CR>'//HY1_V1316A4/PV_D' = '_EDC_VLV_PV:CLOSED';	STOP	NO



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Module	Name	Description	Expression	Force SP State	Delay On (s)
HY1_PU1282B	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_V1381BD/PV_D' = 'EDC_VLV_PV:OPEN' AND <CR>('HY1_LSL1282BP/DI1/DISC_ACT.CV' != TRUE AND<CR>'HY1_LS12382B/DI1/DISC_ACT.CV' != TRUE AND<CR>'HY1_LSH1282C1/DI1/DISC_ACT.CV' != TRUE));	START	5
HY1_PU1282B	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_V1381BD/PV_D' = 'EDC_VLV_PV:CLOSED' OR<CR>'HY1_LSL1282BP/DI1/DISC_ACT.CV' = TRUE OR<CR>'HY1_LS12382B/DI1/DISC_ACT.CV' = TRUE OR<CR>'HY1_LSH1282C1/DI1/DISC_ACT.CV' = TRUE));	STOP	NO
HY1_AG1182B	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'HY1_LS12382B/DI1/DISC_ACT.CV' != TRUE;	START	NO
HY1_AG1182B	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'HY1_LS12382B/DI1/DISC_ACT.CV' != TRUE;	STOP	NO
HY1_PU2135B	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_V1235/PV_D' = 'EDC_VLV_PV:OPEN' AND <CR>'HY1_V1235BD/PV_D' = 'EDC_VLV_PV:OPEN')	START	5
HY1_PU2135B	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_V1235/PV_D' != 'EDC_VLV_PV:OPEN' OR <CR>'HY1_V1235BD/PV_D' != 'EDC_VLV_PV:OPEN')	STOP	NO
HY1_AG1135B	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND ('HY1_LSL1135B/DI1/DISC_ACT' != TRUE)	START	NO
HY1_AG1135B	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND ('HY1_LSL1135B/DI1/DISC_ACT' = TRUE)	STOP	NO
HY1_PU1135A	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_V1235/PV_D' = 'EDC_VLV_PV:OPEN' AND <CR>'HY1_V1135/PV_D' = 'EDC_VLV_PV:OPEN')	START	5



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HY1_PU1135A	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_V1235/PV_D' != '_EDC_VLV_PV:OPEN' OR <CR>'HY1_V1135/PV_D' != '_EDC_VLV_PV:OPEN')	STOP	NO
HY1_AG1135A	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND ('HY1_LSL1135A/DI1/DISC_ACT' != TRUE)	START	NO
HY1_AG1135A	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND ('HY1_LSL1135A/DI1/DISC_ACT' = TRUE)	STOP	NO
HY1_PU1078TW	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_LSL1078TW/DI1/DISC_ACT.CV' != TRUE)	START	NO
HY1_PU1078TW	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('HY1_LSL1078TW/DI1/DISC_ACT.CV' = TRUE)	STOP	NO
HY1_TT1281A	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND<CR>('PV.CV' <= 'REQ_SP.CV' AND <CR>'MODELOCK_OVR.CV' = FALSE)	0	NO
HY1_FT1282B	Condition 1	Auto Logic	('TP01.CV' = TRUE AND<CR>'MODELOCK_OVR.CV' = FALSE AND<CR>('HY1_V1381BD/PV_D' = '_EDC_VLV_PV:CLOSED' OR<CR>'HY1_LSL1282BP/DI1/DISC_ACT.CV' = TRUE OR<CR>'HY1_LS12382B/DI1/DISC_ACT.CV' = TRUE OR<CR>'HY1_LSH1282C1/DI1/DISC_ACT.CV' = TRUE OR<CR>'HY1_PU1282B/PV_D' = '_EDC_MTR2_PV:STOPPED'));	0	NO
HY1_FT1203H	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND ('MODELOCK_OVR.CV' = FALSE) AND <CR>('HY1_PT1203/PID1/HI_HI_ACT.CV' = TRUE) OR <CR>('HY1_PSH1203/DI1/DISC_ACT' = TRUE) OR <CR>('HY1_RD12031/DI1/DISC_ACT.CV' = TRUE) OR <CR>('HY1_V1203F/PV_D' != '_EDC_VLV_PV:CLOSED') OR <CR>('HY1_V1203D/PV_D' != '_EDC_VLV_PV:CLOSED') OR <CR>('HY1_V1203VA/PV_D' != '_EDC_VLV_PV:CLOSED') OR <CR>('HY1_V1203V1/PV_D' != '_EDC_VLV_PV:CLOSED') OR ('HY1_V1203H/PV_D' = '_EDC_VLV_PV:CLOSED'))	0	NO



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PT. Wilmar Nabati Indonesia - DCS Serang Project

3258585

Module	Name	Description	Expression	Force SP State	Delay On (s)
HY1_FT1001	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND ('MODELOCK_OVR.CV' = FALSE) AND <CR>('HY1_V1001F/PV_D' = '_EDC_VLV_PV:CLOSED' or <CR>'HY1_V1002F/PV_D' = '_EDC_VLV_PV:CLOSED');	0	NO

Table 69: Module Force Setpoints